

# INVERSE TRIGONOMETRY FUNCTION

## JEE EXERCISES

### Exercise J-1

#### BASIC INTRODUCTION

1. If  $\sum_{i=1}^{20} \sin^{-1} x_i = 10\pi$ , then  $\sum_{i=1}^{20} x_i$  is equal to :

- (A) 20 (B) 10 (C) 0 (D) None of these

यदि  $\sum_{i=1}^{20} \sin^{-1} x_i = 10\pi$ , तो  $\sum_{i=1}^{20} x_i$  का मान है :

- (A) 20 (B) 10 (C) 0 (D) इनमें से कोई नहीं

**Ans.** A

**Sol.**  $\sin^{-1} x_1 + \sin^{-1} x_2 + \sin^{-1} x_3 + \dots + \sin^{-1} x_{20} = 10\pi$

$\downarrow \quad \downarrow \quad \downarrow \quad \quad \downarrow$

$$\frac{\pi}{2} \quad \frac{\pi}{2} \quad \frac{\pi}{2} \quad \quad \frac{\pi}{2}$$

$$x_1 = 1 \quad x_2 = 1 \quad x_3 = 1 \quad \quad x_{20} = 1$$

$$x_1 + x_2 + \dots + x_{20} = 1 + 1 + 1 + \dots + 1$$

$$= 20$$

2. If  $\cos^{-1}(a) + \cos^{-1}(b) + \cos^{-1}(c) = 3\pi$  and  $f(1) = 2$ ,  $f(x + y) = f(x) f(y)$  for all  $x, y$ ; then

$a^{2f(1)} + b^{2f(2)} + c^{2f(3)} + \frac{(a+b+c)}{a^{2f(1)} + b^{2f(2)} + c^{2f(3)}}$  is equal to :

यदि  $\cos^{-1}(a) + \cos^{-1}(b) + \cos^{-1}(c) = 3\pi$  तथा  $f(1) = 2$  एवं सभी  $x, y$  के लिए  $f(x + y) = f(x) f(y)$  हो तो

$a^{2f(1)} + b^{2f(2)} + c^{2f(3)} + \frac{(a+b+c)}{a^{2f(1)} + b^{2f(2)} + c^{2f(3)}}$  का मान है :

- (A) 0 (B) 1 (C) 2 (D) 3

**Ans.** C

**Sol.**  $\cos^{-1}(a) + \cos^{-1}(b) + \cos^{-1}(c) = 3\pi$

$\downarrow \quad \downarrow \quad \downarrow$

$$\pi \quad \pi \quad \pi$$

$$f(x + y) = f(x) f(y)$$

$$f(1) = 2$$

$$f(1+1) = f(1) \times f(1)$$

$$f(2) = (2)(2) = 4$$

$$f(2+1) = f(2) f(1)$$

$$f(3) = 4 \times 2 = 8$$

$$\Rightarrow a^{2f(1)} + b^{2f(2)} + c^{2f(3)} + \frac{(a+b+c)}{a^{2f(1)} + b^{2f(2)} + c^{2f(3)}}$$

$$\Rightarrow (-1)^4 + (-1)^8 + (-1)^{16} + \frac{(-1-1-1)}{(-1)^4 + (-1)^8 + (-1)^{16}}$$

$$\Rightarrow 1+1+1 + \frac{-3}{1+1+1}$$

$$= 3 - \frac{3}{3} = 3 - 1 = 2$$

### DOMAIN AND RANGE OF ITFs

3. If  $\sin^{-1}x + \tan^{-1}x = y$  ( $-1 < x < 1$ ), then which is not possible :

यदि  $\sin^{-1}x + \tan^{-1}x = y$  ( $-1 < x < 1$ ), तो कौनसा सम्भव नहीं है :

(A)  $y = \frac{3\pi}{2}$

(B)  $y = 0$

(C)  $y = \frac{\pi}{2}$

(D)  $y = -\frac{\pi}{2}$

**Ans.** A

**Sol.**  $\sin^{-1}x + \tan^{-1}x = y$  ( $-1 < x < 1$ )

$$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right) + \left(-\frac{\pi}{4}, \frac{\pi}{4}\right) = y$$

$$\left(-\frac{3\pi}{4}, \frac{3\pi}{4}\right) = y$$

$$y = \frac{3\pi}{2}$$

## EVALUATION

4. The value of  $\sin^2\left(\cos^{-1}\frac{1}{2}\right) + \cos^2\left(\sin^{-1}\frac{1}{3}\right)$  is :

- (A)  $\frac{17}{36}$  (B)  $\frac{59}{36}$  (C)  $\frac{36}{59}$  (D) None of these

$\sin^2\left(\cos^{-1}\frac{1}{2}\right) + \cos^2\left(\sin^{-1}\frac{1}{3}\right)$  का मान है :

- (A)  $\frac{17}{36}$  (B)  $\frac{59}{36}$  (C)  $\frac{36}{59}$  (D) इनमें से कोई नहीं

**Ans.** B

**Sol.** 
$$\begin{aligned} & \sin^2\left(\cos^{-1}\frac{1}{2}\right) + \cos^2\left(\sin^{-1}\frac{1}{3}\right) \\ &= \sin^2 \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) + \cos^2\left(\cos^{-1}\left(\frac{\sqrt{8}}{3}\right)\right) \\ &\Rightarrow \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{\sqrt{8}}{3}\right)^2 \\ &= \frac{3}{4} + \frac{8}{9} \\ &= \frac{27+32}{36} = \frac{59}{36} \end{aligned}$$

5.  $\cot\left[\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{8}\right] =$

- (A) 1 (B) -1 (C)  $\sqrt{2}$  (D)  $-\sqrt{2}$

**Ans.** A

**Sol.** 
$$\cot\left[\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{8}\right]$$

$$\cot \left[ \tan^{-1} \left( \frac{\frac{1}{2} + \frac{1}{8}}{1 - \frac{1}{16}} \right) + \tan^{-1} \frac{1}{5} \right]$$

$$\cot \left[ \tan^{-1} \left( \frac{2}{3} \right) + \tan^{-1} \frac{1}{5} \right]$$

$$\cot \left[ \tan^{-1} \left( \frac{\frac{2}{3} + \frac{1}{5}}{1 - \frac{2}{15}} \right) \right]$$

$$\cot \left[ \tan^{-1} \left( \frac{10+3}{13} \right) \right]$$

$$\cot (\tan^{-1} (1))$$

$$= 1$$

6.  $\cot \left[ \tan^{-1} \left( \frac{1}{7} \right) + \tan^{-1} \left( \frac{1}{13} \right) \right]$  is equal to :

$\cot \left[ \tan^{-1} \left( \frac{1}{7} \right) + \tan^{-1} \left( \frac{1}{13} \right) \right]$  का मान है :

(A)  $\frac{2}{9}$

(B)  $\frac{9}{2}$

(C)  $\frac{7}{9}$

(D)  $\frac{9}{7}$

**Ans.** B

**Sol.**  $\cot \left[ \tan^{-1} \left( \frac{1}{7} \right) + \tan^{-1} \left( \frac{1}{13} \right) \right]$

$$\cot \left[ \tan^{-1} \left( \frac{\frac{1}{7} + \frac{1}{13}}{1 - \frac{1}{91}} \right) \right]$$

$$\cot \left[ \tan^{-1} \left( \frac{13+7}{91-1} \right) \right]$$

$$\cot \left[ \tan^{-1} \left( \frac{20}{91} \right) \right]$$

$$\cot \left( \cot^{-1} \left( \frac{9}{2} \right) \right)$$

$$= \frac{9}{2}$$

### PROPERTIES OF ITF

7. If  $0 < x < 1$ , then  $\sqrt{1+x^2} [\{x \cos (\cot^{-1} x) + \sin (\cot^{-1} x)\}^2 - 1]^{1/2} =$  [JEE-2008]

यदि  $0 < x < 1$ , तो  $\sqrt{1+x^2} [\{x \cos (\cot^{-1} x) + \sin (\cot^{-1} x)\}^2 - 1]^{1/2} =$  [JEE-2008]

- (A)  $\frac{x}{\sqrt{1+x^2}}$  (B)  $x$  (C)  $x\sqrt{1+x^2}$  (D)  $\sqrt{1+x^2}$

Ans. C

Sol.  $\sqrt{1+x^2} [\{x \cos (\cot^{-1} x) + \sin (\cot^{-1} x)\}^2 - 1]^{1/2}$

$$\sqrt{1+x^2} \left[ \left\{ x \frac{x}{\sqrt{x^2+1}} + \frac{1}{\sqrt{x^2+1}} \right\}^2 - 1 \right]^{1/2}$$

$$\sqrt{1+x^2} \left[ \frac{(x^2+1)^2}{x^2+1} - 1 \right]^{1/2}$$

$$\sqrt{1+x^2} (x)$$

### OTHER PROPERTIES OF ITF

8. If  $a \leq \tan^{-1}x + \cot^{-1}x + \sin^{-1}x \leq b$ . Then :

- (A)  $a = 0, b = \pi$  (B)  $b = \frac{\pi}{2}$  (C)  $a = \frac{\pi}{4}$  (D) None of these

यदि  $a \leq \tan^{-1}x + \cot^{-1}x + \sin^{-1}x \leq b$ . तब :

- (A)  $a = 0, b = \pi$  (B)  $b = \frac{\pi}{2}$  (C)  $a = \frac{\pi}{4}$  (D) इनमें से कोई नहीं

**Ans.** A

**Sol.**  $a \leq \underbrace{\tan^{-1} x + \cot^{-1} x + \sin^{-1} x}_y \leq b$

$$x \in [-1, 1]$$

$$y = \left[ \frac{-\pi}{4}, \frac{\pi}{4} \right] + \left[ \frac{\pi}{4}, \frac{3\pi}{4} \right] + \left[ \frac{-\pi}{2}, \frac{\pi}{2} \right]$$

$$y_{\min} = \frac{-\pi}{4} + \frac{3\pi}{4} - \frac{\pi}{2} \Rightarrow a = 0$$

$$y_{\max} = \frac{\pi}{4} + \frac{\pi}{4} + \frac{\pi}{2} = \pi \Rightarrow b = \pi$$

9.  $\sin^{-1}x + \sin^{-1}\frac{1}{x} + \cos^{-1}x + \cos^{-1}\frac{1}{x} =$

(A)  $\pi$  (B)  $\frac{\pi}{2}$  (C)  $\frac{3\pi}{2}$  (D) None of these

$$\sin^{-1}x + \sin^{-1}\frac{1}{x} + \cos^{-1}x + \cos^{-1}\frac{1}{x} =$$

(A)  $\pi$  (B)  $\frac{\pi}{2}$  (C)  $\frac{3\pi}{2}$  (D) इनमें से कोई नहीं

**Ans.** A

**Sol.**  $\sin^{-1}x + \sin^{-1}\frac{1}{x} + \cos^{-1}x + \cos^{-1}\frac{1}{x}$

$$\sin^{-1}x + \cos^{-1}x + \operatorname{cosec}^{-1}x + \sec^{-1}x$$

$$= \frac{\pi}{2} + \frac{\pi}{2} = \pi$$

10. If  $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$ , then  $\cos^{-1}x + \cos^{-1}y =$

$$\text{यदि } \sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}, \text{ तो } \cos^{-1}x + \cos^{-1}y =$$

(A)  $\frac{2\pi}{3}$  (B)  $\frac{\pi}{3}$  (C)  $\frac{\pi}{6}$  (D)  $\pi$

**Ans.** B

**Sol.**  $\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$

$$\frac{\pi}{2} - \cos^{-1} x + \frac{\pi}{2} - \cos^{-1} y = \frac{2\pi}{3}$$

$$\cos^{-1} x + \cos^{-1} y = \pi - \frac{2\pi}{3} = \frac{\pi}{3}$$

11.  $\sin \left[ 2 \cos^{-1} \left( -\frac{3}{5} \right) \right]$  is equal to :

$\sin \left[ 2 \cos^{-1} \left( -\frac{3}{5} \right) \right]$  का मान है :

(A)  $\frac{6}{25}$

(B)  $\frac{24}{25}$

(C)  $\frac{4}{5}$

(D)  $-\frac{24}{25}$

**Ans.** D

**Sol.**  $\sin \left[ 2 \cos^{-1} \left( -\frac{3}{5} \right) \right]$

$$\cos^{-1} \left( \frac{-3}{5} \right) = A$$

$$\pi + \cos^{-1} \left( \frac{3}{5} \right) = A$$

$$-\frac{3}{5} = \cos A$$

$$\sin 2A = 2 \sin A \cos A$$

$$= 2 \left( \frac{4}{5} \right) \left( -\frac{3}{5} \right) = -\frac{24}{25}$$

### ADDITION OR SUBTRACTION OF TWO OR MORE ANGLES

12. If  $\cos^{-1} \frac{3}{5} - \sin^{-1} \frac{4}{5} = \cos^{-1} x$ , then  $x =$

यदि  $\cos^{-1} \frac{3}{5} - \sin^{-1} \frac{4}{5} = \cos^{-1} x$ , तो  $x =$

- (A) 0                      (B) 1                      (C)  $\frac{1}{2}$                       (D)  $\frac{1}{4}$

**Ans.** B

**Sol.**  $\cos^{-1} \frac{3}{5} - \sin^{-1} \frac{4}{5} = \cos^{-1} x$

$$\cos^{-1} \left( \frac{3}{5} \right) - \cos^{-1} \left( \frac{3}{1} \right) = \cos^{-1} x$$

$$\cos^{-1} x = 0$$

$$x = 1$$

13. If  $\sin^{-1} \frac{1}{3} + \sin^{-1} \frac{2}{3} = \sin^{-1} x$ , then  $x$  is equal to :

यदि  $\sin^{-1} \frac{1}{3} + \sin^{-1} \frac{2}{3} = \sin^{-1} x$ , तो  $x$  बराबर है :

- (A) 0                      (B)  $\frac{\sqrt{5}-4\sqrt{2}}{9}$                       (C)  $\frac{\sqrt{5}+4\sqrt{2}}{9}$                       (D)  $\frac{\pi}{2}$

**Ans.** C

**Sol.**  $\sin^{-1} \frac{1}{3} + \sin^{-1} \frac{2}{3} = \sin^{-1} x$

$$\sin A = \frac{1}{3} \quad \sin B = \frac{2}{3}$$

$$\cos A = \frac{\sqrt{8}}{3} \quad \cos B = \frac{\sqrt{5}}{3}$$

$$A + B = \sin^{-1} x$$

$$\sin(A + B) = x$$

$$x = \sin A \cos B + \cos A \sin B$$

$$= \frac{1}{3} \frac{\sqrt{5}}{3} + \frac{\sqrt{8}}{3} \frac{2}{3}$$

$$x = \frac{\sqrt{5} + 4\sqrt{2}}{9}$$



14. If  $\theta = \cot^{-1} \sqrt{\cos x} - \tan^{-1} \sqrt{\cos x}$ , then  $\sin \theta =$

- (A)  $\tan \frac{1}{2} x$  (B)  $\tan^2 (x/2)$  (C)  $\frac{1}{2} \tan^{-1}(x/2)$  (D) None of these

यदि  $\theta = \cot^{-1} \sqrt{\cos x} - \tan^{-1} \sqrt{\cos x}$ , तब  $\sin \theta =$

- (A)  $\tan \frac{1}{2} x$  (B)  $\tan^2 (x/2)$  (C)  $\frac{1}{2} \tan^{-1}(x/2)$  (D) इनमें से कोई नहीं

**Ans.** B

**Sol.** Given that  $\cot^{-1} (\sqrt{\cos \alpha}) - \tan^{-1} (\sqrt{\cos \alpha}) = x$  .....(i)

We know that,  $\cot^{-1} (\sqrt{\cos \alpha}) + \tan^{-1} (\sqrt{\cos \alpha}) = \frac{\pi}{2}$  .....(ii)

On adding equations (i) and (ii), we get

$$2 \cot^{-1} (\sqrt{\cos \alpha}) = \frac{\pi}{2} + x \Rightarrow \sqrt{\cos \alpha} = \cot \left( \frac{\pi}{4} + \frac{x}{2} \right) \Rightarrow \sqrt{\cos \alpha} = \frac{\cot \frac{x}{2} - 1}{1 + \cot \frac{x}{2}}$$

$$\Rightarrow \sqrt{\cos \alpha} = \frac{\cos \frac{x}{2} - \sin \frac{x}{2}}{\cos \frac{x}{2} + \sin \frac{x}{2}} \Rightarrow \cos \alpha = \frac{1 - \sin x}{1 + \sin x} \Rightarrow \frac{1 - \tan^2 \frac{\alpha}{2}}{1 + \tan^2 \frac{\alpha}{2}} = \frac{1 - \sin x}{1 + \sin x}$$

Applying componendo and dividendo rule, we get  $\sin x = \tan^2 \left( \frac{\alpha}{2} \right)$

15. If a, b, c be positive real numbers and the value of

$\theta = \tan^{-1} \sqrt{\frac{a(a+b+c)}{bc}} + \tan^{-1} \sqrt{\frac{b(a+b+c)}{ca}} + \tan^{-1} \sqrt{\frac{c(a+b+c)}{ab}}$  then  $\tan \theta$  is equal to :

- (A) 0 (B) 1 (C)  $\frac{a+b+c}{abc}$  (D) None of these

यदि a, b, c धनात्मक वास्तविक संख्याएँ हैं तथा  $\theta = \tan^{-1} \sqrt{\frac{a(a+b+c)}{bc}} + \tan^{-1} \sqrt{\frac{b(a+b+c)}{ca}} + \tan^{-1} \sqrt{\frac{c(a+b+c)}{ab}}$  तो  $\tan \theta$  बराबर है :

- (A) 0 (B) 1 (C)  $\frac{a+b+c}{abc}$  (D) इनमें से कोई नहीं

**Ans.** A

**Sol.**  $\theta = \tan^{-1} \sqrt{\frac{a(a+b+c)}{bc}} + \tan^{-1} \sqrt{\frac{b(a+b+c)}{ca}} + \tan^{-1} \sqrt{\frac{c(a+b+c)}{ab}}$

$\downarrow$                        $\downarrow$                        $\downarrow$   
 A                      B                      C

$$\theta = A + B + C$$

$$\tan \theta = \tan (A + B + C)$$

$$\Rightarrow \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$$

$$\tan \theta = \frac{\sqrt{\frac{a(a+b+c)}{bc}} + \sqrt{\frac{b(a+b+c)}{ca}} + \sqrt{\frac{c(a+b+c)}{ab}} - \frac{(a+b+c)}{\sqrt{abc}}}{1 - \frac{a+b+c}{c} - \frac{a+b+c}{b} - \frac{a+b+c}{a}}$$

$$\Rightarrow \tan \theta = 0$$

### ITF EQUATIONS

16. If  $3 \cos^{-1}(x^2 - 7x + 25/2) = \pi$ , then  $x =$
- (A) Only 3                      (B) Only 4                      (C) 3 or 4                      (D) None of these

- यदि  $3 \cos^{-1}(x^2 - 7x + 25/2) = \pi$ , तो  $x =$
- (A) केवल 3                      (B) केवल 4                      (C) 3 or 4                      (D) इनमें से कोई नहीं

**Ans.** C

**Sol.**  $3 \cos^{-1}(x^2 - 7x + 25/2) = \pi$

$$\cos^{-1}\left(x^2 - 7x + \frac{25}{2}\right) = \frac{\pi}{3}$$

$$x^2 - 7x + \frac{25}{2} = \frac{1}{2}$$

$$2x^2 - 14x + 25 = 1$$

$$2x^2 - 14x + 24 = 0 \quad x = 3$$

$$x^2 - 7x + 12 = 0 \quad \Rightarrow 4 - 21 + \frac{3}{2} = \frac{-x_1 + 25}{2} = \frac{1}{2}$$

$$x = 3, 4$$

$$x = 4$$

17. If  $\tan(x + y) = 33$  and  $x = \tan^{-1} 3$ , then  $y$  will be :

यदि  $\tan(x + y) = 33$  तथा  $x = \tan^{-1} 3$ , तो  $y$  होगा :

- (A) 0.3                      (B)  $\tan^{-1}(1.3)$                       (C)  $\tan^{-1}(0.3)$                       (D)  $\tan^{-1}\left(\frac{1}{18}\right)$

**Ans.** C

**Sol.**  $\tan(x + y) = 33$                        $x = \tan^{-1} = 3$   
 $\tan x = 3$

$$\frac{\tan x + \tan y}{1 + \tan x \tan y} = 33$$

$$\frac{3 + t}{1 - 3t} = 33$$

$$3 + t = 33 - 99t$$

$$100 \tan y = 30$$

$$\tan y = 0.3$$

$$y = \tan^{-1}(0.3)$$

18. If  $(\tan^{-1}x)^2 + (\cot^{-1}x)^2 = \frac{5\pi^2}{8}$ , then x equals :

- (A) -1                      (B) 1                      (C) 0                      (D) None of these

यदि  $(\tan^{-1}x)^2 + (\cot^{-1}x)^2 = \frac{5\pi^2}{8}$ , तो x बराबर है :

- (A) -1                      (B) 1                      (C) 0                      (D) इनमें से कोई नहीं

**Ans.** A

**Sol.**  $(\tan^{-1}x)^2 + (\cot^{-1}x)^2 = \frac{5\pi^2}{8}$

$$t^2 + \left(\frac{\pi}{2} - t\right)^2 = \frac{5\pi^2}{8}$$

$$t^2 + \frac{\pi^2}{4} + t^2 - \pi t = \frac{5\pi^2}{8}$$

$$2t^2 - \pi t - \frac{3\pi^2}{8} = 0$$

$$16t^2 - 8\pi t - 3\pi^2 = 0$$

$$16t^2 - 12\pi t + 4\pi t - 3\pi^2 = 0$$

$$(4 \tan^{-1} x - 3\pi)(4 \tan^{-1} x + \pi) = 0$$

$$\Rightarrow 4t(4t - 3\pi) + \pi(4t - 3\pi) = 0$$

$$\tan^{-1} x = \frac{3\pi}{4} \quad \tan^{-1} x = \frac{-\pi}{4}$$

(rejected)  $x = -1$

19. If  $\tan^{-1} \frac{1}{a-1} = \tan^{-1} \frac{1}{x} + \tan^{-1} \frac{1}{a^2-x+1}$ , then x is :

यदि  $\tan^{-1} \frac{1}{a-1} = \tan^{-1} \frac{1}{x} + \tan^{-1} \frac{1}{a^2-x+1}$ , तो x बराबर है :

- (A)  $\frac{a}{2}$  (B)  $a^3$  (C)  $a^2 - a + 1$  (D)  $a^2 + a - 1$

**Ans.** C

**Sol.**  $\tan^{-1} \frac{1}{a-1} = \tan^{-1} \frac{1}{x} + \tan^{-1} \frac{1}{a^2-x+1}$

$$\frac{1}{a-1} = \frac{\frac{1}{x} + \frac{1}{a^2-x+1}}{1 - \frac{1}{x(a^2-x+1)}}$$

$$\frac{1}{a-1} = \frac{a^2-x+1+x}{x(a^2-x+1)-1}$$

$$xa^2 - x^2 + x - 1 = (a-1)(a^2+1)$$

$$xa^2 - x^2 + x - 1 = a^3 + a - a^2 - 1$$

$$x^2 - (a^2+1)x + a^3 - a^2 + a = 0$$

$$x = \frac{(a^2+1) \pm \sqrt{a^4 - 4a^3 + 6a^2 - 4a + 1}}{2}$$

$$= \frac{(a^2+1) \pm \sqrt{(a-1)^4}}{2} = \frac{(a^2+1) \pm (a^2-2a+1)}{2}$$

$$x = \frac{a^2+1+a^2-2a+1}{2}, \frac{a^2+1-a^2+2a-1}{2}$$

$$x = a^2 - a + 1, a$$

20. The trigonometric equation,  $\sin^{-1} x = 2 \sin^{-1} a$  has a solution for :

- (A) all real values of a (B)  $|a| < \frac{1}{2}$  (C)  $|a| \leq \frac{1}{\sqrt{2}}$  (D)  $\frac{1}{2} < |a| < \frac{1}{\sqrt{2}}$

त्रिकोणमितीय समीकरण  $\sin^{-1} x = 2 \sin^{-1} a$  का एक हल सम्भव होगा यदि :

- (A) a के सभी मान के लिए (B)  $|a| < \frac{1}{2}$  (C)  $|a| \leq \frac{1}{\sqrt{2}}$  (D)  $\frac{1}{2} < |a| < \frac{1}{\sqrt{2}}$

**Ans.** C

**Sol.**  $-\frac{\pi}{2} \leq 2 \sin^{-1} a \leq \frac{\pi}{2}$

$$-\frac{\pi}{4} \leq \sin^{-1} a \leq \frac{\pi}{4}$$

$$-\frac{1}{\sqrt{2}} \leq a \leq \frac{1}{\sqrt{2}}$$

$$|a| \leq \frac{1}{\sqrt{2}}$$

21. If  $\cos^{-1} x - \cos^{-1} \left( \frac{y}{2} \right) = \alpha$ , then  $4x^2 - 4xy \cos \alpha + y^2$  is equal to :

- (A) 4 (B)  $2 \sin 2\alpha$  (C)  $-4 \sin^2 \alpha$  (D)  $4 \sin^2 \alpha$

यदि  $\cos^{-1} x - \cos^{-1} \left( \frac{y}{2} \right) = \alpha$ , तो  $4x^2 - 4xy \cos \alpha + y^2$  बराबर होगा :

- (A) 4 (B)  $2 \sin 2\alpha$  (C)  $-4 \sin^2 \alpha$  (D)  $4 \sin^2 \alpha$

**Ans.** D

**Sol.**  $\sin^{-1} x = 2 \sin^{-1} a$

$$A - B = \alpha$$

$$\cos A \cos B + \sin A \sin B = \cos \alpha$$

$$xy + \sqrt{1-x^2} \sqrt{4-y^2} = \cos \alpha$$

$$(1-x^2)(4-y^2) = \cos^2 \alpha + x^2 y^2 - 4xy \cos \alpha$$

$$4 - y^2 - 4x^2 + x^2 y^2 = \cos^2 \alpha + x^2 y^2 - 4xy \cos \alpha$$

$$4 - 4\cos^2 \alpha = 4x^2 - 4xy \cos \alpha + y^2$$

$$4(1 - c^2)$$

$$\Rightarrow 4x^2 - 4xy \cos \alpha + y^2 = 4 \sin^2 \alpha$$

## SUMMATION OF SERIES

22. The sum of the infinite terms of the series  $\cot^{-1}\left(1^2 + \frac{3}{4}\right) + \cot^{-1}\left(2^2 + \frac{3}{4}\right) + \cot^{-1}\left(3^2 + \frac{3}{4}\right) + \dots$  is equal to :
- (A)  $\tan^{-1}(1)$                       (B)  $\tan^{-1}(2)$                       (C)  $\tan^{-1}(3)$                       (D)  $\tan^{-1}(4)$

अनन्त श्रेणी  $\cot^{-1}\left(1^2 + \frac{3}{4}\right) + \cot^{-1}\left(2^2 + \frac{3}{4}\right) + \cot^{-1}\left(3^2 + \frac{3}{4}\right) + \dots$  का योग होगा :

- (A)  $\tan^{-1}(1)$                       (B)  $\tan^{-1}(2)$                       (C)  $\tan^{-1}(3)$                       (D)  $\tan^{-1}(4)$

**Ans.** B

**Sol.**  $T_n = \cot^{-1}\left(n^2 + \frac{3}{4}\right) = \tan^{-1}\left(\frac{1}{n^2 + (3/4)}\right) = \tan^{-1}\left(\frac{1}{1 + n^2 - (1/4)}\right)$

$$= \tan^{-1}\left(\frac{1}{1 + \left(n - \frac{1}{2}\right)\left(n + \frac{1}{2}\right)}\right) = \tan^{-1}\left(\frac{\left(n + \frac{1}{2}\right) - \left(n - \frac{1}{2}\right)}{1 + \left(n + \frac{1}{2}\right)\left(n - \frac{1}{2}\right)}\right)$$

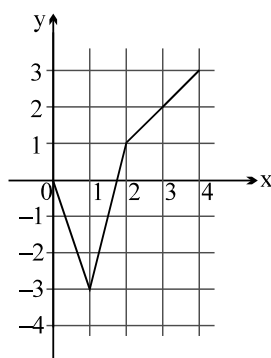
## Exercise J-2

### BASIC INTRODUCTION

1. Given is a partial graph of an even periodic function  $f$  whose period is 8. If  $[ ]$  denotes greatest integer function then find the value of the expression.

दिया गया ग्राफ एक सम व आवर्ती फलन  $f$  का है जिसका आवर्तकाल 8 है।  $[*]$  महत्तम पूर्णांक फलन को निरूपित करता है। निम्न का मान ज्ञात कीजिए।

$$f(-3) + 2 |f(-1)| + \left[ f\left(\frac{7}{8}\right) \right] + f(0) + \arccos(f(-2)) + f(-7) + f(20)$$



**Ans.** 5

**Sol.** Even function

Periodic with period = 8

$$f(-x) = f(x)$$

$$f(-3) + 2 |f(-1)| + \left[ f\left(\frac{7}{8}\right) \right] + f(0) + \cos^{-1}(f(-2)) + f(-7) + f(20)$$

$$= 2 + 2(3) + (-3) + 0 + \cos^{-1}(1) + (-3) + 3$$

$$= 2 + 6 - 3 + 0$$

$$= 5$$

### DOMAIN AND RANGE BASED QUESTIONS

2. Find the domain of definition the following functions. (Read the symbols  $[ ]$  and  $\{ \}$  as greatest integers and fractional part functions respectively.)

निम्नलिखित फलनों का प्रान्त ज्ञात कीजिये। ( $[*]$  व  $\{*\}$  महत्तम पूर्णांक फलन व भिन्नात्मक भाग फलन है)

(i)  $f(x) = \arccos \frac{2x}{1+x}$

$$(ii) f(x) = \sqrt{\cos(\sin x)} + \sin^{-1} \frac{1+x^2}{2x}$$

$$(iii) f(x) = \sin^{-1} \left( \frac{x-3}{2} \right) - \log_{10}(4-x)$$

$$(iv) f(x) = \sin^{-1}(2x+x^2)$$

$$(v) f(x) = \frac{\sqrt{1-\sin x}}{\log_5(1-4x^2)} + \cos^{-1}(1-\{x\})$$

$$(vi) f(x) = \sqrt{3-x} + \cos^{-1} \left( \frac{3-2x}{5} \right) + \log_6(2|x|-3) + \sin^{-1}(\log_2 x)$$

$$(vii) f(x) = \log_{10}(1 - \log_7(x^2 - 5x + 13)) + \cos^{-1} \left( \frac{3}{2 + \sin \frac{9\pi x}{2}} \right)$$

$$(viii) f(x) = e^{\sin^{-1}(\frac{x}{2})} + \tan^{-1} \left[ \frac{x}{2} - 1 \right] + \ln(\sqrt{x} - [x])$$

**Ans.** (i)  $-1/3 \leq x \leq 1$  (ii)  $\{1, -1\}$  (iii)  $1 \leq x < 4$  (iv)  $[-(1 + \sqrt{2}), (\sqrt{2}, -1)]$   
(v)  $x \in (-1/2, 1/2), x \neq 0$  (vi)  $(3/2, 2]$  (vii)  $\{7/3, 25/9\}$   
(viii)  $(-2, 2) - \{-1, 0, 1\}$  (ix)  $\{x \mid x = 2n\pi + \frac{\pi}{6}, n \in \mathbb{I}\}$

**Sol.** (i)  $f(x) = \cos^{-1} \left( \frac{2x}{1+x} \right)$

$$-1 \leq \frac{2x}{1+x} \leq 1$$

$$\frac{2x}{x+1} + 1 \geq 0 \quad \frac{2x}{x+1} - 1 \leq 0$$

$$\frac{2x+x+1}{x+1} \geq 0 \quad \frac{2x-x-1}{x+1} \leq 0$$

$$x \in \left[ -\frac{1}{3}, 1 \right]$$

$$(ii) f(x) = \sqrt{\cos(\sin x)} + \sin^{-1} \left( \frac{1+x^2}{2x} \right)$$

$$-1 \leq \frac{1+x^2}{2x} \leq 1$$



$$\frac{1+x^2}{2x} + 1 \leq 0$$

$$\frac{1+x^2}{2x} - 1 \leq 0$$

$$\frac{x^2+2x+1}{2x} \geq 0$$

$$\frac{x^2-2x+1}{2x} \leq 0$$

$$\frac{(x+1)^2}{2x} \geq 0$$

$$\frac{(x-1)^2}{2x} \leq 0$$

$$x = -1, 1$$

$$(iii) f(x) = \sin^{-1}\left(\frac{x-3}{2}\right) - \log_{10}(4-x)$$

$$-1 \leq \frac{x-3}{2} \leq 1$$

$$x \in [1, 4]$$

$$(iv) f(x) = \sin^{-1}(2x + x^2)$$

$$-1 \leq x^2 + 2x \leq 1$$

$$x^2 + 2x + 1 \geq 0$$

$$x^2 + 2x - 1 \leq 0$$

$$x \in \left[-(1+\sqrt{2}), \sqrt{2}-1\right]$$

$$(v) f(x) = \frac{\sqrt{1-\sin x}}{\log_5(1-4x^2)} + \cos^{-1}(1-\{x\})$$

$$1-4x^2 > 0$$

$$1-4x^2 \neq 1 \Rightarrow x \neq 0$$

$$(2x-1)(2x+1) < 0$$

$$x \in 0$$

$$x \in \left(-\frac{1}{2}, \frac{1}{2}\right) - \{0\}$$

$$(vi) f(x) = \sqrt{3-x} + \cos^{-1}\left(\frac{3-2x}{5}\right) + \log_6(2|x|-3) + \sin^{-1}(\log_2 x)$$

$$3-2 \geq 0$$

$$-1 \leq \frac{3-2x}{5} \leq 1$$

$$x \leq 3$$

$$-5 \leq 3-2x$$

$$3-2x \leq 5$$

$$x \leq 4$$

$$x \geq -1$$

$$x \in [-1, 4]$$

$$2|x|-3 > 0$$

$$x \in \left(-\infty, -\frac{3}{2}\right) \cup \left(\frac{3}{2}, \infty\right)$$

$$-1 \leq \log_2 x \leq 1$$

$$\log_2 x \geq -1 \quad \log_2 x \leq 1$$

$$x \in \left[\frac{1}{2}, 2\right] \quad \text{f.a.} \Rightarrow x \in \left(\frac{3}{2}, 2\right]$$

$$(vii) (1 - \log_7 (x^2 - 5x + 13)) > 0$$

$$\log_7 (x^2 - 5x + 13) < 1$$

$$x^2 - 5x + 13 < 7$$

$$x^2 - 5x + 6 < 0$$

$$(x - 3)(x - 2) < 0$$

$$x \in (2, 3)$$

$$x^2 - 5x + 13 > 0$$

$$x \in \mathbb{R}$$

$$-1 \leq \frac{3}{2 + \sin \frac{9\pi x}{2}} \leq 1$$

$$x = \left(\frac{4n+1}{9}\right)$$

$$(viii) f(x) = e^{\sin^{-1}(\frac{x}{2})} + \tan^{-1}\left[\frac{x}{2} - 1\right] + \ln(\sqrt{x - [x]})$$

$$-1 \leq \frac{x}{2} \leq 1 \quad x \in \mathbb{R}$$

$$x \in [-2, 2]$$

$$x \in (-2, 2) - \{-1, 0, 1\}$$

3. Identify the pair(s) of functions which are identical.

निम्न युग्मों में समरूप फलन ज्ञात कीजिये व ग्राफ बनाईये।

$$(a) y = \tan(\cos^{-1} x); y = \frac{\sqrt{1-x^2}}{x}$$

$$(b) y = \tan(\cot^{-1} x); y = \frac{1}{x}$$

$$(c) y = \sin(\arctan x); y = \frac{x}{\sqrt{1+x^2}}$$

$$(d) y = \cos(\arctan x); y = \sin(\operatorname{arccot} x)$$

**Ans.** (a), (b), (c) and (d) all are identical

**Sol.** (a)  $y = \tan(\cos^{-1} x)$

$$= \tan \left( \tan^{-1} \left( \frac{\sqrt{1-x^2}}{x} \right) \right) = \frac{\sqrt{1-x^2}}{x}$$

$$(b) y = \tan (\cot^{-1} x) = \tan \left( \tan^{-1} \frac{1}{x} \right) = \frac{1}{x}$$

$$(c) y = \sin (\tan^{-1} x) = \sin \left( \sin^{-1} \frac{x}{\sqrt{1+x^2}} \right) = \frac{x}{\sqrt{1+x^2}}$$

$$(d) y = \cos (\tan^{-1} x) \qquad y = \sin \left( \cot^{-1} \frac{x}{1} \right)$$

$$= \cos \left( \cos^{-1} \left( \frac{1}{\sqrt{1+x^2}} \right) \right) = \sin \left( \sin^{-1} \frac{1}{\sqrt{1+x^2}} \right)$$

$$= \frac{1}{\sqrt{1+x^2}} \qquad = \frac{1}{\sqrt{1+x^2}}$$

4. Find the domain and range of the following functions. (Read the symbols  $[]$  and  $\{\}$  as greatest integers and fractional part functions respectively.)

निम्नलिखित फलनों का प्रान्त व परिसर ज्ञात कीजिए। ( $[*]$  व  $\{*\}$  महत्तम पूर्णांक फलन व अंशिक फलन निम्न है)

$$(i) f(x) = \cot^{-1}(2x-x^2)$$

$$(ii) f(x) = \sec^{-1}(\log_3 \tan x + \log_{\tan x} 3)$$

$$(iii) f(x) = \cos^{-1} \left( \frac{\sqrt{2x^2+1}}{x^2+1} \right)$$

$$(iv) f(x) = \tan^{-1} \left( \log_{\frac{4}{5}} (5x^2 - 8x + 4) \right)$$

**Ans.** (i)  $D : x \in \mathbb{R} \quad R : [\pi/4, \pi)$

$$(ii) D: x \in \left( n\pi, n\pi + \frac{\pi}{2} \right) - \left\{ x \mid x = n\pi + \frac{\pi}{4} \right\} \quad n \in \mathbb{I}; \quad R : \left[ \frac{\pi}{3}, \frac{2\pi}{3} \right] - \left[ \frac{\pi}{2} \right]$$

$$(iii) D : x \in \mathbb{R} \quad R : \left[ 0, \frac{\pi}{2} \right)$$

$$(iv) D : x \in \mathbb{R} \quad R : \left( -\frac{\pi}{2}, \frac{\pi}{4} \right]$$

**Sol.** (i)  $f(x) = \cot^{-1}(2x-x^2)$

$$2x - x^2 \in \mathbb{R}$$

$$-x^2 + 2x$$

$$\left(-\infty, -\frac{D}{4a}\right]$$

$$\left(-\infty, +\frac{4}{4(+1)}\right] \Rightarrow (-\infty, 1]$$

$$\text{Rang} \Rightarrow \left[\frac{\pi}{4}, \pi\right)$$

$$(ii) f(x) = \sec^{-1}(\log_3 \tan x + \log_{\tan x} 3)$$

$$|\log_3 \tan x + \log_{\tan x} 3| \geq -1$$

$$\left|\log_3 \tan x + \frac{1}{\log_3 \tan x}\right| \geq 2$$

$$\tan x > 0 \quad \tan x \neq 1$$

$$x \in \left(n\pi, n\pi + \frac{\pi}{2}\right) - \left\{n\pi + \frac{\pi}{4}\right\}$$

$$\text{Range } \sin^{-1} [(-\infty, -2] \cup [2, \infty)]$$

$$\left[\frac{\pi}{3}, \frac{2\pi}{3}\right] - \left[\frac{\pi}{2}\right]$$

$$(iii) \quad f(x) = \cos^{-1} \left( \frac{\sqrt{2x^2 + 1}}{x^2 + 1} \right)$$

$$-1 \leq \frac{\sqrt{2x^2 + 1}}{x^2 + 1} \leq 1$$

$$\frac{\sqrt{2x^2 + 1}}{x^2 + 1} + 1 \geq 0$$

$$\frac{\sqrt{2x^2 + 1}}{x^2 + 1} - 1 \leq 0$$

$$\frac{\sqrt{2x^2 + 1} + x^2 + 1}{x^2 + 1} \geq 0$$

$$\frac{\sqrt{2x^2 + 1} + x^2 + 1}{x^2 + 1} \leq 0$$

$$x \in \mathbb{R}$$

$$\sqrt{2x^2 + 1} \leq x^2 + 1$$

$$x^2 \geq 0$$

$$x \in \mathbb{R}$$

$$\text{Domain} \Rightarrow x \in \mathbb{R}$$

$$\text{Range} \Rightarrow \left[0, \frac{\pi}{2}\right)$$

$$(iv) \quad f(x) = \tan^{-1} \left( \log_{\frac{4}{5}} (5x^2 - 8x + 4) \right)$$

5. Let  $f(x) = \cot^{-1}(x^2 + 4x + \alpha^2 - \alpha)$  be a function defined  $\mathbb{R} \rightarrow \left(0, \frac{\pi}{2}\right]$  then find the complete set of real values of  $\alpha$  for which  $f(x)$  is onto.

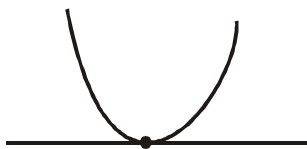
$$f(x) = \cot^{-1}(x^2 + 4x + \alpha^2 - \alpha) \text{ फलन } \mathbb{R} \rightarrow \left(0, \frac{\pi}{2}\right] \text{ में परिभाषित है तो } \alpha \text{ के वह सभी मान ज्ञात कीजिए जिसके लिये } f(x)$$

आच्छादक है।

**Ans.**  $\frac{1 \pm \sqrt{17}}{2}$

**Sol.**  $f(x) = \cot^{-1}(x^2 + 4x + \alpha^2 - \alpha) \quad \mathbb{R} \rightarrow \left(0, \frac{\pi}{2}\right]$

$$[0, \infty)$$



$$D = 0$$

$$16 - 4(\alpha^2 - \alpha) = 0$$

$$\alpha^2 - \alpha - 4 = 0$$

$$\alpha = \frac{1 \pm \sqrt{17}}{2}$$

6. If the range of the function  $\cos^{-1}|1-x^2|$  is  $\left[a, \frac{\pi}{b}\right]$  then find the value of b.

यदि फलन  $\cos^{-1}|1-x^2|$  का परिसर  $\left[a, \frac{\pi}{b}\right]$  है, तो b का मान ज्ञात कीजिए।

**Ans.** 2

**Sol.**  $\cos^{-1}|1-x^2|$

$$-1 \leq |1-x^2| \leq 1$$

$$0 \leq |1-x^2| \leq 1$$

$$\cos^{-1} 0 = \frac{\pi}{2} \quad \cos^{-1}(1) = 0$$

$$\left[0, \frac{\pi}{2}\right]$$

### EVALUATION

7. (a) Find the following

(a) निम्नलिखित के मान ज्ञात कीजिये।

(i)  $\tan\left[\cos^{-1}\frac{1}{2} + \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)\right]$

(ii)  $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$

(iii)  $\cos\left(\tan^{-1}\frac{3}{4}\right)$

(iv)  $\tan\left(\sin^{-1}\frac{3}{5} + \cot^{-1}\frac{3}{2}\right)$

- (b) Find the following :

निम्नलिखित के मान ज्ञात कीजिये :

(i)  $\sin\left[\frac{\pi}{2} - \sin^{-1}\left(\frac{-\sqrt{3}}{2}\right)\right]$

(ii)  $\cos\left[\cos^{-1}\left(\frac{-\sqrt{3}}{2}\right) + \frac{\pi}{6}\right]$

(iii)  $\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$

(iv)  $\sin\left(\frac{1}{4}\arcsin\frac{\sqrt{63}}{8}\right)$

**Ans.** (a) (i)  $\frac{1}{\sqrt{3}}$ , (ii)  $\frac{5\pi}{6}$ , (iii)  $\frac{4}{5}$ , (iv)  $\frac{17}{6}$

(b) (i)  $\frac{1}{2}$ , (ii)  $-1$ , (iii)  $-\frac{\pi}{4}$ , (iv)  $\frac{\sqrt{2}}{4}$

**Sol.** (i)  $\tan\left[\cos^{-1}\frac{1}{2} + \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)\right]$

$$\tan\left[\tan^{-1}\sqrt{3} - \tan^{-1}\frac{1}{\sqrt{3}}\right]$$

$$\tan\left(\frac{\pi}{3} - \frac{\pi}{6}\right)$$

$$\tan\frac{\pi}{6} = \left(\frac{1}{\sqrt{3}}\right)$$

(ii)  $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$

$$\pi - \frac{\pi}{6} \Rightarrow \frac{5\pi}{6}$$

(iii)  $\cos\left(\tan^{-1}\frac{3}{4}\right) = \cos\left(\cos^{-1}\frac{4}{5}\right) = \frac{4}{5}$

(iv)  $\tan\left(\sin^{-1}\frac{3}{5} + \cot^{-1}\frac{3}{2}\right)$

$$\tan\left[\tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{2}{3}\right)\right]$$

$$\tan\left[\tan^{-1}\frac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{1}{2}}\right]$$

$$= \tan\left[\tan^{-1}\left(\frac{9+8}{12 \times \frac{1}{2}}\right)\right] = \frac{17}{6}$$

(b) (i)  $\sin\left[\frac{\pi}{2} - \sin^{-1}\left(\frac{-\sqrt{3}}{2}\right)\right]$

$$= \sin \left[ \frac{\pi}{2} + \frac{\pi}{3} \right] = \cos \frac{\pi}{3} = \frac{1}{2}$$

$$(ii) \quad \cos \left[ \cos^{-1} \left( \frac{-\sqrt{3}}{2} \right) + \frac{\pi}{6} \right]$$

$$= \cos \left[ \frac{5\pi}{6} + \frac{\pi}{6} \right] = \cos \pi = (-1)$$

$$(iii) \quad \tan^{-1} \left( \tan \frac{3\pi}{4} \right) = -\frac{\pi}{4}$$

$$(iv) \quad \sin \left( \frac{1}{4} \arcsin \frac{\sqrt{63}}{8} \right)$$

$$\sin^{-1} \frac{\sqrt{63}}{8} = \theta \Rightarrow \sin \theta = \frac{\sqrt{63}}{8}$$

$$\sin \frac{\theta}{4} = ?$$

$$\sin \theta = \frac{\sqrt{63}}{8}$$

$$\cos \theta = \frac{1}{8}$$

$$\cos \frac{\theta}{2} = \sqrt{\frac{1 + \cos \theta}{2}} = \frac{3}{4}$$

$$\sin \frac{\theta}{4} = \sqrt{\frac{1 - \cos \frac{\theta}{2}}{2}} = \sqrt{\frac{1 - \frac{3}{4}}{2}}$$

$$\Rightarrow \frac{1}{\sqrt{8}} = \frac{\sqrt{2}}{4}$$

### PROPERTIES OF ITF

8. Show that :  $\sin^{-1} \left( \sin \frac{33\pi}{7} \right) + \cos^{-1} \left( \cos \frac{46\pi}{7} \right) + \tan^{-1} \left( -\tan \frac{13\pi}{8} \right) + \cot^{-1} \left( \cot \left( -\frac{19\pi}{8} \right) \right) = \frac{13\pi}{7}$

प्रदर्शित कीजिए :  $\sin^{-1} \left( \sin \frac{33\pi}{7} \right) + \cos^{-1} \left( \cos \frac{46\pi}{7} \right) + \tan^{-1} \left( -\tan \frac{13\pi}{8} \right) + \cot^{-1} \left( \cot \left( -\frac{19\pi}{8} \right) \right) = \frac{13\pi}{7}$



**Ans.**

**Sol.**  $\sin^{-1}\left(\sin\frac{33\pi}{7}\right) + \cos^{-1}\left(\cos\frac{46\pi}{7}\right) + \tan^{-1}\left(-\tan\frac{13\pi}{8}\right) + \cot^{-1}\left(\cot\left(-\frac{19\pi}{8}\right)\right)$

$$\text{LHS } 5\pi - \frac{33\pi}{7} + \frac{46\pi}{7} - 6\pi - \frac{3\pi}{8} + \pi - \left(\frac{19\pi}{8} - 2\pi\right)$$

$$= \frac{13\pi}{7}$$

9. Find the simplest value of:

मान ज्ञात कीजिये :

(a)  $f(x) = \arccos x + \arccos\left(\frac{x}{2} + \frac{1}{2}\sqrt{3-3x^2}\right)$ ,  $x \in \left(\frac{1}{2}, 1\right)$

(b)  $f(x) = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ ,  $x \in \mathbb{R} - \{0\}$

**Ans.** (a)  $\frac{\pi}{3}$ ; (b)  $\frac{1}{2}\tan^{-1}x$

**Sol.**  $y = \cos^{-1}x + \cos^{-1}\left(\frac{x}{2} + \frac{1}{2}\sqrt{3-3x^2}\right)$ ,  $x \in \left(\frac{1}{2}, 1\right)$

$$\cos^{-1}x = \theta \quad \theta \in \left(0, \frac{\pi}{3}\right)$$

$$\cos\theta = x$$

$$\Rightarrow \theta + \cos^{-1}\left(\frac{\cos\theta}{2} + \frac{\sqrt{3}}{2}\sqrt{2\cos\theta}\right)$$

$$\theta + \cos^{-1}\left(\cos\left(\theta - \frac{\pi}{3}\right)\right) \quad 0 < \theta < \frac{\pi}{3}$$

$$= \theta - \theta + \frac{\pi}{3} = \frac{\pi}{3}$$

(b)  $f(x) = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$   $x \in \mathbb{R} - \{0\}$

$$\Rightarrow x = \tan\theta$$

$$= \tan^{-1} \left( \frac{\sec \theta - 1}{\tan \theta} \right)$$

$$= \tan^{-1} \left( \frac{1 - \cos \theta}{\sin \theta} \right)$$

$$\Rightarrow \tan^{-1} \left( \tan \left( \frac{\theta}{2} \right) \right)$$

$$\Rightarrow \frac{\theta}{2} = \frac{\tan^{-1} x}{2}$$

10. Evaluate :  $\left| \tan^{-1} \left( \frac{x-y}{1+xy} \right) + \tan^{-1} \left( \frac{y-z}{1+yz} \right) + \tan^{-1} \left( \frac{z-x}{1+zx} \right) \right|$

(i) when x, y, z all are of same sign.

(ii) When not all three are of the same sign and (one positive  $\times$  one negative)  $< -1$  and (one positive  $\times$  one negative)  $> -1$ .

हल कीजिए :  $\left| \tan^{-1} \left( \frac{x-y}{1+xy} \right) + \tan^{-1} \left( \frac{y-z}{1+yz} \right) + \tan^{-1} \left( \frac{z-x}{1+zx} \right) \right|$

(i) जब x, y, z सभी के समान चिन्ह हो।

(ii) जब सभी के समान समान चिन्ह नहीं हो (एक धनात्मक  $\times$  एक ऋणात्मक)  $< -1$  तथा (एक धनात्मक  $\times$  एक ऋणात्मक)  $> -1$

**Ans.** (i) 0 (ii)  $\pi$

**Sol.**  $\left| \tan^{-1} \left( \frac{x-y}{1+xy} \right) + \tan^{-1} \left( \frac{y-z}{1+yz} \right) + \tan^{-1} \left( \frac{z-x}{1+zx} \right) \right|$

(i)  $|\tan^{-1}(x) - \tan^{-1}y + \tan^{-1}y - \tan^{-1}z + \tan^{-1}z - \tan^{-1}x|$

$= 0$

(ii)  $x, y > 0 \quad z < 0$

$x(z) < -1 \quad y(z) > -1$

$z = -t$

$xt > 1 \quad yt < 1$

$\Rightarrow \left| \tan^{-1} \left( \frac{x-y}{1+xy} \right) + \tan^{-1} \left( \frac{y+t}{1-yt} \right) + \tan^{-1} \left( \frac{-t-x}{1-tx} \right) \right|$

$$\left| \tan^{-1} x - \tan^{-1} y + \tan^{-1} y + \tan^{-1} t - \left( \pi + \tan^{-1} \left( \frac{t+x}{1-tx} \right) \right) \right|$$

$$\Rightarrow |-\pi|$$

$$= \pi$$

## ADDITION OR SUBTRACTION OF TWO OR MORE ANGLES

11. Prove that :  $\sin \cot^{-1} \tan \cos^{-1} x = \sin \operatorname{cosec}^{-1} \cot \tan^{-1} x = x$  where  $x \in (0, 1]$

सिद्ध कीजिय :  $\sin \cot^{-1} \tan \cos^{-1} x = \sin \operatorname{cosec}^{-1} \cot \tan^{-1} x = x$  where  $x \in (0, 1]$

**Ans.**

**Sol.**  $\sin \left( \cot^{-1} \left( \tan \left( \cos^{-1} \frac{x}{1} \right) \right) \right)$

$$\Rightarrow \sin \left( \cot^{-1} \left( \tan \left( \tan^{-1} \frac{\sqrt{1-x^2}}{x} \right) \right) \right)$$

$$\Rightarrow \sin \left( \cot^{-1} \left( \frac{\sqrt{1-x^2}}{x} \right) \right)$$

$$\Rightarrow \sin \left( \sin^{-1} \left( \frac{x}{1} \right) \right)$$

$$= x$$

$$\sin \operatorname{cosec}^{-1} \left( \cot \left( \cot^{-1} \left( \frac{1}{x} \right) \right) \right)$$

$$\Rightarrow \sin \left( \operatorname{cosec}^{-1} \left( \cot^{-1} \left( \frac{1}{x} \right) \right) \right)$$

$$\Rightarrow \sin \left( \operatorname{cosec}^{-1} \left( \frac{1}{x} \right) \right)$$

$$\Rightarrow \sin (\sin^{-1} (x)) = x$$

12. Prove the identities :

निम्न का सत्यापन कीजिये :

$$(a) \sin^{-1} \cos (\sin^{-1} x) + \cos^{-1} \sin (\cos^{-1} x) = \frac{\pi}{2}, |x| \leq 1$$

$$(b) 2 \tan^{-1} (\operatorname{cosec} \tan^{-1} x - \tan \cot^{-1} x) = \tan^{-1} x \quad (x \neq 0)$$

$$(c) \tan (\tan^{-1} x + \tan^{-1} y + \tan^{-1} z) = \cot (\cot^{-1} x + \cot^{-1} y + \cot^{-1} z)$$

**Ans.**

**Sol.** (a)  $\sin^{-1} \cos (\sin^{-1} x) + \cos^{-1} \sin (\cos^{-1} x) = \frac{\pi}{2}$

LHS

$$\Rightarrow \sin^{-1} (\sqrt{1-x^2}) + \cos^{-1} (\sqrt{1-x^2})$$

$$\Rightarrow \frac{\pi}{2}$$

(b)  $2 \tan^{-1} (\operatorname{cosec} \tan^{-1} x - \tan \cot^{-1} x) = \tan^{-1} x$

$$\text{LHS } 2 \tan^{-1} \left( \operatorname{cosec} \left( \operatorname{cosec}^{-1} \frac{\sqrt{1+x^2}}{x} \right) - \tan \tan^{-1} \frac{1}{x} \right)$$

$$\Rightarrow 2 \tan^{-1} \left( \frac{\sqrt{1+x^2}}{x} - \frac{1}{x} \right)$$

$$\Rightarrow \tan^{-1} \left( \frac{2 \left( \frac{\sqrt{1+x^2}-1}{x} \right)}{1 - \frac{(\sqrt{1+x^2}-1)^2}{x^2}} \right) = \tan^{-1} \left( \frac{2(\sqrt{1+x^2}-1)}{\frac{x^2-1-x^2-1+2\sqrt{1+x^2}}{x^2}} \right) \Rightarrow \tan^{-1}(x)$$

(c)  $\tan (\tan^{-1} x + \tan^{-1} y + \tan^{-1} z) = \cot (\cot^{-1} x + \cot^{-1} y + \cot^{-1} z)$

$$\text{LHS } \Rightarrow \tan \left[ \frac{\pi}{2} - \cot^{-1} x + \frac{\pi}{2} - \cot^{-1} y + \frac{\pi}{2} - \cot^{-1} z \right]$$

$$\tan \left[ \frac{3\pi}{2} - (\cot^{-1} x + \cot^{-1} y + \cot^{-1} z) \right]$$

$$\Rightarrow \cot(\cot^{-1} x + \cot^{-1} y + \cot^{-1} z)$$

### ADDITION OR SUBTRACTION OF TWO OR MORE ANGLES

13. Let  $\alpha = \sin^{-1}\left(\frac{36}{85}\right)$ ,  $\beta = \cos^{-1}\left(\frac{4}{5}\right)$  and  $\gamma = \tan^{-1}\left(\frac{8}{15}\right)$ , find  $(\alpha + \beta + \gamma)$  and hence prove that

यदि  $\alpha = \sin^{-1}\left(\frac{36}{85}\right)$ ,  $\beta = \cos^{-1}\left(\frac{4}{5}\right)$  और  $\gamma = \tan^{-1}\left(\frac{8}{15}\right)$ ,  $(\alpha + \beta + \gamma)$  ज्ञात कीजिए और सिद्ध कीजिए

$$(i) \sum \cot \alpha = \prod \cot \alpha, \quad (ii) \sum \tan \alpha \cdot \tan \beta = 1$$

**Ans.**  $\pi/2$

**Sol.** (i)  $\alpha = \sin^{-1}\left(\frac{36}{85}\right)$ ,  $\beta = \cos^{-1}\left(\frac{4}{5}\right)$  and  $\gamma = \tan^{-1}\left(\frac{8}{15}\right)$

$$\alpha + \beta + \gamma = \sin^{-1}\left(\frac{36}{85}\right) + \cos^{-1}\left(\frac{4}{5}\right) + \tan^{-1}\left(\frac{8}{15}\right)$$

$$= \tan^{-1}\left(\frac{36}{77}\right) + \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{8}{15}\right)$$

$$= \tan^{-1}\left(\frac{36}{77}\right) + \tan^{-1}\left(\frac{\frac{3}{4} + \frac{8}{15}}{1 - \frac{24}{60}}\right)$$

$$= \tan^{-1}\left(\frac{36}{77}\right) + \tan^{-1}\left(\frac{45 + 32}{60 - 24}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{36}{77}\right) + \tan^{-1}\left(\frac{77}{36}\right)$$

$$\Rightarrow \frac{\pi}{2}$$

(ii)  $\cot \alpha + \cot \beta + \cot \gamma = \cot \alpha \cot \beta \cot \gamma$

$$\frac{1}{\tan \alpha} + \frac{1}{\tan \beta} + \frac{1}{\tan \gamma} = \frac{1}{\tan \alpha \tan \beta \tan \gamma}$$

$$\tan \alpha \tan \beta + \tan \beta \tan \gamma + \tan \gamma \tan \alpha = 1$$

$$\alpha + \beta + \gamma = \frac{\pi}{2}$$

$$\tan(\alpha + \beta + \gamma) : \tan\left(\frac{\pi}{2}\right)$$

$$(ii) \sum \tan \alpha \cdot \tan \beta = 1$$

14. Prove that :

सिद्ध कीजिय :

$$(a) 2 \cos^{-1} \frac{3}{\sqrt{13}} + \cot^{-1} \frac{16}{63} + \frac{1}{2} \cos^{-1} \frac{7}{25} = \pi$$

$$(b) \cos^{-1}\left(\frac{5}{13}\right) + \cos^{-1}\left(-\frac{7}{25}\right) + \sin^{-1} \frac{36}{325} = \pi$$

$$(c) \arccos \sqrt{\frac{2}{3}} - \arccos \frac{\sqrt{6} + 1}{2\sqrt{3}} = \frac{\pi}{6}$$

**Ans.**

$$\text{Sol. } (a) 2 \cos^{-1} \frac{3}{\sqrt{13}} + \cot^{-1} \frac{16}{63} + \frac{1}{2} \cos^{-1} \frac{7}{25} = \pi$$

A

B

C

$$2 \cos^{-1}\left(\frac{3}{\sqrt{13}}\right) = A$$

$$\cos \frac{A}{2} = \frac{3}{\sqrt{13}}$$

$$\Rightarrow \tan \frac{A}{2} = \frac{2}{3}$$

$$\tan A = \frac{2 \tan \frac{A}{2}}{1 - \tan^2 \frac{A}{2}} = \frac{2 \times \frac{2}{3}}{1 - \frac{4}{9}} = \frac{\frac{4}{3}}{\frac{5}{9}} = \frac{12}{5}$$

$$\tan A = \frac{12}{5} \Rightarrow A = \tan^{-1}\left(\frac{12}{5}\right)$$

$$\cot^{-1}\left(\frac{16}{63}\right) = B$$

$$\cot^{-1} B = \frac{16}{63}$$

$$\tan B = \frac{63}{16} \Rightarrow B = \tan^{-1} \left( \frac{63}{16} \right)$$

$$\frac{1}{2} \cos^{-1} \frac{7}{25} = C \Rightarrow \cos 2C = \frac{7}{25}$$

$$\frac{1 - \tan^2 C}{1 + \tan^2 C} = \frac{7}{25}$$

$$\Rightarrow 25 - 25 \tan^2 C = 7 + 7 \tan^2 C$$

$$\tan^2 C = \frac{9}{16} \Rightarrow \tan C = \frac{3}{4} \Rightarrow C = \tan^{-1} \left( \frac{3}{4} \right)$$

$$\Rightarrow A + B + C$$

$$= \tan^{-1} \left( \frac{12}{5} \right) + \tan^{-1} \left( \frac{63}{16} \right) + \tan^{-1} \left( \frac{3}{4} \right)$$

$$\Rightarrow \tan^{-1} \left( \frac{12}{5} \right) + \tan^{-1} \left( \frac{3}{4} \right) + \tan^{-1} \left( \frac{63}{16} \right)$$

$$= \pi + \tan^{-1} \left( \frac{\frac{12}{5} + \frac{3}{4}}{1 - \frac{12}{5} \times \frac{3}{4}} \right) + \tan^{-1} \left( \frac{63}{16} \right)$$

$$= \pi + \tan^{-1} \left( \frac{48 + 15}{20 - 36} \right) + \tan^{-1} \left( \frac{63}{16} \right)$$

$$\Rightarrow \pi + \tan^{-1} \left( \frac{-63}{16} \right) + \tan^{-1} \left( \frac{63}{16} \right)$$

$$\Rightarrow \pi$$

$$(b) \cos^{-1} \left( \frac{5}{13} \right) + \cos^{-1} \left( -\frac{7}{25} \right) + \sin^{-1} \frac{36}{325} = \pi$$

$$\Rightarrow \tan^{-1} \left( \frac{12}{5} \right) + \pi - \tan^{-1} \left( \frac{24}{7} \right) + \tan^{-1} \left( \frac{36}{323} \right)$$

$$\Rightarrow \pi + \tan^{-1} \left( \frac{\frac{12}{5} - \frac{24}{7}}{1 + \frac{12}{5} \times \frac{24}{7}} \right) + \tan^{-1} \left( \frac{36}{323} \right)$$

$$\Rightarrow \pi + \tan^{-1} \left( \frac{84-120}{35+288} \right) + \tan^{-1} \left( \frac{36}{323} \right)$$

$$\Rightarrow \pi + \tan^{-1} \left( -\frac{36}{323} \right) + \tan^{-1} \left( \frac{36}{323} \right)$$

$$\Rightarrow \pi$$

$$(c) \cos \sqrt{\frac{2}{3}} - \arccos \frac{\sqrt{6}+1}{2\sqrt{3}} = \frac{\pi}{6}$$

$$\text{LHS} \Rightarrow \tan^{-1} \frac{1}{\sqrt{2}} - \tan^{-1} \left( \frac{\sqrt{3}-\sqrt{2}}{\sqrt{6}+1} \right)$$

$$\Rightarrow \tan^{-1} \frac{1}{\sqrt{2}} - \left[ \tan^{-1} (\sqrt{3}) - \tan^{-1} \sqrt{2} \right]$$

$$\Rightarrow \cot^{-1} \sqrt{2} - \tan^{-1} \sqrt{3} + \tan^{-1} \sqrt{2}$$

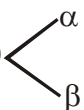
$$\Rightarrow \frac{\pi}{2} - \frac{\pi}{3}$$

$$= \frac{\pi}{6}$$

15. If  $\alpha$  and  $\beta$  are the roots of the equation  $x^2 + 5x - 49 = 0$  then find the value of  $\cot(\cot^{-1}\alpha + \cot^{-1}\beta)$ .

यदि  $\alpha$  व  $\beta$  समीकरण  $x^2 + 5x - 49 = 0$  के मूल हैं तो  $\cot(\cot^{-1}\alpha + \cot^{-1}\beta)$  का मान ज्ञात कीजिए।

**Ans.** 10

**Sol.**  $x^2 + 5x - 49 = 0$  

$$\alpha + \beta = -5 \quad \alpha\beta = -49$$

$$\cot(\cot^{-1}\alpha + \cot^{-1}\beta)$$

$$\cot A = \alpha \quad \cot B = \beta$$

$$\cot(A+B) = \frac{\cot A \cot B - 1}{\cot A + \cot B}$$

$$\Rightarrow \frac{\alpha\beta - 1}{\alpha + \beta} = \frac{-49 - 1}{-5} = \frac{-50}{-5} = 10$$



16. If  $a > b > c > 0$  then find the value of:  $\cot^{-1}\left(\frac{ab+1}{a-b}\right) + \cot^{-1}\left(\frac{bc+1}{b-c}\right) + \cot^{-1}\left(\frac{ca+1}{c-a}\right)$ .

यदि  $a > b > c > 0$  तो मान ज्ञात कीजिये :  $\cot^{-1}\left(\frac{ab+1}{a-b}\right) + \cot^{-1}\left(\frac{bc+1}{b-c}\right) + \cot^{-1}\left(\frac{ca+1}{c-a}\right)$

**Ans.**  $\pi$

**Sol.**  $\cot^{-1}\left(\frac{ab+1}{a-b}\right) + \cot^{-1}\left(\frac{bc+1}{b-c}\right) + \cot^{-1}\left(\frac{ca+1}{c-a}\right)$

$$\tan^{-1}\left(\frac{a-b}{1+ab}\right) + \tan^{-1}\left(\frac{b-c}{1+bc}\right) + \tan^{-1}\left(\frac{c-a}{1+ca}\right)$$

$$\Rightarrow \tan^{-1} a - \tan^{-1} b + \tan^{-1} b - \tan^{-1} c + \pi + \tan^{-1} c - \tan^{-1} a$$

$$= \pi$$

### ITF EQUATIONS

17. Prove that :  $\tan^{-1} n + \cot^{-1}(n+1) = \tan^{-1}(n^2 + n + 1)$ .

सिद्ध करो कि :  $\tan^{-1} n + \cot^{-1}(n+1) = \tan^{-1}(n^2 + n + 1)$ .

**Ans.**

**Sol.**  $\tan^{-1} n + \cot^{-1}(n+1) = \tan^{-1}(n^2 + n + 1)$

$$\text{LHS} \Rightarrow \tan^{-1}(n) + \tan^{-1}\left(\frac{1}{n+1}\right)$$

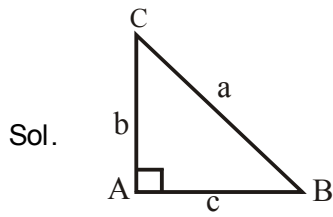
$$\Rightarrow \tan^{-1}\left(\frac{n + \frac{1}{n+1}}{1 - \frac{n}{n+1}}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{n^2 + n + 1}{n + 1 - n}\right) = \tan^{-1}(n^2 + n + 1)$$

18. In a triangle ABC if  $\angle A = 90^\circ$ , then prove that,  $\tan^{-1} \frac{b}{c+a} + \tan^{-1} \frac{c}{a+b} = \frac{\pi}{4}$ .

यदि त्रिभुज ABC में  $\angle A = 90^\circ$  तब सिद्ध कीजिए कि  $\tan^{-1} \frac{b}{c+a} + \tan^{-1} \frac{c}{a+b} = \frac{\pi}{4}$

**Ans.**



$$a^2 = b^2 + c^2$$

$$\Rightarrow \tan^{-1}\left(\frac{b}{a+c}\right) + \tan^{-1}\left(\frac{c}{a+b}\right) = \frac{\pi}{4}$$

$$\text{LHS } \tan^{-1}\left(\frac{\frac{b}{a+c} + \frac{c}{a+b}}{1 - \frac{bc}{(a+b)(a+c)}}\right)$$

$$\Rightarrow \tan^{-1}\left[\frac{ab + b^2 + ac + c^2}{a^2 + ac + ab + bc - bc}\right] = \tan^{-1}\left[\frac{ab + ac + a^2}{a^2 + ac + ab}\right]$$

$$\Rightarrow \frac{\pi}{4}$$

19. Solve for x and y :  $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$  and  $\cos^{-1}x - \cos^{-1}y = \frac{\pi}{3}$ .

x व y के लिए समीकरण  $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$  एवं  $\cos^{-1}x - \cos^{-1}y = \frac{\pi}{3}$  को हल कीजिए।

**Ans.**  $x = \frac{1}{2}, y = 1$

**Sol.**  $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$

$$\cos^{-1}x - \cos^{-1}y = \frac{\pi}{3}$$

$$\frac{\pi}{2} - \sin^{-1}x - \frac{\pi}{2} + \sin^{-1}y = \frac{\pi}{3}$$

$$-\sin^{-1} x + \sin^{-1} y = \frac{\pi}{3}$$

$$2 \sin^{-1} = \pi$$

$$\sin^{-1} = \frac{\pi}{2} \Rightarrow y = 1$$

$$\sin^{-1} x + \frac{\pi}{2} = \frac{2\pi}{3}$$

$$\sin^{-1} x = \frac{2\pi}{3} - \frac{\pi}{2} = \frac{\pi}{6}$$

$$x = \frac{1}{2}$$

20. Find all possible value of 'p' and 'q' for which  $\cos^{-1} \sqrt{p} + \cos^{-1} \sqrt{1-p} + \cos^{-1} \sqrt{1-q} = \frac{3\pi}{4}$ .

'p' व 'q' के सभी सम्भव मान ज्ञात करो जो कि समीकरण  $\cos^{-1} \sqrt{p} + \cos^{-1} \sqrt{1-p} + \cos^{-1} \sqrt{1-q} = \frac{3\pi}{4}$

**Ans.**  $0 \leq p \leq 1$  and  $q = \frac{1}{2}$

**Sol.**  $\cos^{-1} \sqrt{p} + \cos^{-1} \sqrt{1-p} + \cos^{-1} \sqrt{1-q} = \frac{3\pi}{4}$

$$\tan^{-1} \frac{\sqrt{1-p}}{\sqrt{p}} + \tan^{-1} \frac{\sqrt{p}}{\sqrt{1-p}} + \cos^{-1} \sqrt{1-q} = \frac{3\pi}{4}$$

$$\tan^{-1} \frac{\sqrt{1-p}}{\sqrt{p}} + \cot^{-1} \frac{\sqrt{1-p}}{\sqrt{p}} + \cos^{-1} \sqrt{1-q} = \frac{3\pi}{4}$$

$$\frac{\pi}{2} + \cos^{-1} \sqrt{1-q} = \frac{3\pi}{4}$$

$$\cos^{-1} \sqrt{1-q} = \frac{\pi}{4}$$

$$\sqrt{1-q} = \frac{1}{\sqrt{2}}$$

$$1-q = \frac{1}{2}$$

$$q = \frac{1}{2}$$

$$p \in [0, 1]$$

21. Solve the following equations :

निम्नलिखित को हल कीजिये :

$$(a) \tan^{-1}(x-1) + \tan^{-1}(x) + \tan^{-1}(x+1) = \tan^{-1}(3x)$$

$$(b) \tan^{-1} \frac{x-1}{x+1} + \tan^{-1} \frac{2x-1}{2x+1} = \tan^{-1} \frac{23}{36}$$

$$(c) \cos^{-1} \frac{x^2-1}{x^2+1} + \tan^{-1} \frac{2x}{x^2-1} = \frac{2\pi}{3}$$

**Ans.** (a)  $x = 0, \frac{1}{2}, -\frac{1}{2}$  (b)  $x = \frac{4}{3}$  (c)  $x = 2-\sqrt{3}$  or  $\sqrt{3}$

**Sol.** (a)  $\tan^{-1}(x-1) + \tan^{-1}(x) + \tan^{-1}(x+1) = \tan^{-1}(3x)$

$$A \quad B \quad C \quad D$$

$$A + B + C = D$$

$$\tan(A + B + C) = \tan(D)$$

$$\frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A} = 3x$$

$$\frac{3x - x(x^2 - 1)}{1 - x^2 + x - x^2 - x - x^2 + 1} = 3x$$

$$\frac{x(3 - x^2 + 1)}{2 - 3x^2} = 3x$$

$$\Rightarrow x = 0$$

$$4 - x^2 = 6 - 9x^2$$

$$8x^2 = 2$$

$$\Rightarrow x^2 = \frac{1}{4}$$

$$\Rightarrow x = \pm \frac{1}{2}$$

$$(b) \tan^{-1} \frac{x-1}{x+1} + \tan^{-1} \frac{2x-1}{2x+1} = \tan^{-1} \frac{23}{36}$$

$$A + B = C$$

$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = \tan C$$

$$\frac{\frac{x-1}{x+1} + \frac{2x-1}{2x+1}}{1 - \frac{x-1}{x+1} \times \frac{2x-1}{2x+1}} = \frac{23}{36}$$

$$\frac{2x^2 + x - 2x - 1 + 2x^2 + 2x - x - 1}{2x^2 + x + 2x + 1 - 2x^2 + x + 2x - 1} = \frac{23}{36}$$

$$\frac{4x^2 - 2}{6x} = \frac{23}{12}$$

$$24x^2 - 12 = 23x$$

$$24x^2 - 23x - 12 = 0$$

$$24x^2 - 32x + 9x - 12 = 0$$

$$8x(3x - 4) + 3(3x - 4) = 0$$

$$x = \frac{4}{3}$$

$$(c) \cos^{-1} \frac{x^2 - 1}{x^2 + 1} + \tan^{-1} \frac{2x}{x^2 - 1} = \frac{2\pi}{3}$$

$$x = \tan \theta \Rightarrow \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\cos^{-1} \left( -\left( \frac{1 - \tan^{-1} \theta}{1 + \tan^{-1} \theta} \right) \right) + \tan^{-1} \left( -\frac{(2 \tan \theta)}{1 - \tan^{-1} \theta} \right) = \frac{2\pi}{3}$$

$$\pi - \cos^{-1}(\cos 2\theta) - \tan^{-1}(\tan 2\theta) = \frac{2\pi}{3}$$

### SUMMATION OF SERIES

22.  $\tan^{-1} \frac{1}{x^2 + x + 1} + \tan^{-1} \frac{1}{x^2 + 3x + 3} + \tan^{-1} \frac{1}{x^2 + 5x + 7} + \tan^{-1} \frac{1}{x^2 + 7x + 13}$  to n terms, where  $x > 0$ .

$\tan^{-1} \frac{1}{x^2 + x + 1} + \tan^{-1} \frac{1}{x^2 + 3x + 3} + \tan^{-1} \frac{1}{x^2 + 5x + 7} + \tan^{-1} \frac{1}{x^2 + 7x + 13}$  n पदों तक, जहाँ  $x > 0$

**Ans.**  $\tan^{-1}(x+n) - \tan^{-1} x$

**Sol.**  $\tan^{-1}\left(\frac{1}{x^2 + x + 1}\right) + \tan^{-1}\left(\frac{1}{x^2 + 3x + 3}\right) + \tan^{-1}\left(\frac{1}{x^2 + 5x + 7}\right) + \tan^{-1}\left(\frac{1}{x^2 + 7x + 13}\right)$

$$1 + 3 + 5 + \dots = n^2$$

$$S = 1 + 3 + 7 + 13 + \dots T_n$$

$$S = 1 + 3 + 7 + \dots T_{n-1} + T_n$$

$$S - S = 1 + 2 + 4 + 6 + \dots - T_n$$

$$T_n = 1 + \frac{n-1}{2} [2 \times 2 + (n-1)2]$$

$$T_n = n^2 - n + 1$$

$$T_n = \tan^{-1} \left[ \frac{1}{x^2 + (2n-1)x + (n^2 - n + 1)} \right]$$

$$= \tan^{-1} \left[ \frac{1}{1 + x^2 + nx + (n-1)x + n(x-1)} \right]$$

$$= \tan^{-1} \left[ \frac{1}{1 + x(n+x) + (n-1)(n+x)} \right]$$

$$= \tan^{-1} \left[ \frac{1}{1 + (n+x)(n+x-1)} \right]$$

$$= \tan^{-1} \left[ \frac{(n+x) - (x+n-1)}{1 + (n+x)(n+x-1)} \right]$$

$$T_n = \tan^{-1}(n+x) - \tan^{-1}(n+x-1)$$

$$T_1 = \tan^{-1}(1+x) - \tan^{-1}(x)$$

$$T_2 = \tan^{-1}(2+x) - \tan^{-1}(1+x)$$

$$T_3 = \tan^{-1}(3+x) - \tan^{-1}(2+x)$$

$$\vdots \quad \quad \quad \vdots$$

$$T_n = \tan^{-1}(n+x) - \tan^{-1}(n+x-1)$$

$$S_n = \tan^{-1}(n+x) - \tan^{-1}x$$

$$23. \quad \sin^{-1} \frac{1}{\sqrt{2}} + \sin^{-1} \frac{\sqrt{2}-1}{\sqrt{6}} + \dots + \sin^{-1} \frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}} + \dots \infty$$

**Ans.**  $\pi/2$

$$\text{Sol.} \quad \sin^{-1} \frac{1}{\sqrt{2}} + \sin^{-1} \frac{\sqrt{2}-1}{\sqrt{6}} + \dots + \sin^{-1} \frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n(n+1)}} + \dots \infty$$

$$T_n = \sin^{-1} \left( \frac{\sqrt{n}-\sqrt{n-1}}{\sqrt{n}\sqrt{n+1}} \right)$$

$$n^2 + x - n - n + 1 - 2\sqrt{n(n-1)}$$

$$n^2 - n + 1 - 2\sqrt{n(n-1)}$$

$$1 + n(n-1) - 2\sqrt{n(n-1)}$$

$$(n^2 + n) - (n + n - 1 - 2\sqrt{n(n-1)})$$

$$n^2 + n - n - n + 1 + 2\sqrt{n(n-1)}$$

$$\sqrt{n^2 - n + 1 + 2\sqrt{n(n-1)}}$$

$$(n\sqrt{n-1})^2 + 1 + n2\sqrt{n(n-1)}$$

$$\sqrt{n(n-1)} + 1$$

$$S_n = \tan^{-1} \sqrt{n} - 0$$

$$S_n = \tan^{-1} + \infty$$

$$= \frac{\pi}{2}$$

### Paragraph for Question Nos. 24 to 25

Consider the function  $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ ,  $g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$  and  $h(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$ .

माना फलन  $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ ,  $g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$  तथा  $h(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$ .

24. If  $x \in (-1, 1)$ , then the solution of the equation  $f(x) + g(x) + h(x) = \frac{\pi}{2}$ , is equal to :

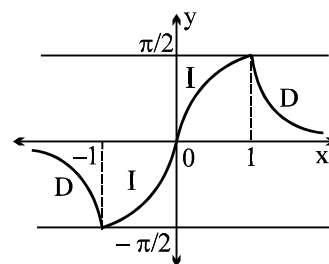
यदि  $x \in (-1, 1)$  है तो समीकरण  $f(x) + g(x) + h(x) = \frac{\pi}{2}$  का हल होगा :

- (A)  $\sqrt{2} - 1$       (B)  $2 - \sqrt{3}$       (C)  $\frac{1}{\sqrt{3}}$       (D)  $\frac{1}{2}$

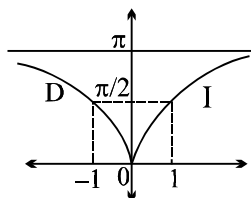
**Ans.** B

**Sol.** starting with  $x = \tan \theta$

$$f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \begin{cases} 2 \tan^{-1} x & -1 \leq x \leq 1 \\ \pi - 2 \tan^{-1} x & \text{if } x > 1 \\ -\pi - 2 \tan^{-1} x & x < -1 \end{cases}$$

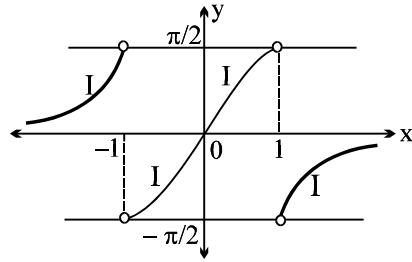


$$g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \begin{cases} 2 \tan^{-1} x & x \geq 0 \\ -2 \tan^{-1} x & x < 0 \end{cases}$$





$$h(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right) = \begin{cases} \pi + 2 \tan^{-1} x & x < -1 \\ 2 \tan^{-1} x & -1 < x < 1 \\ 2 \tan^{-1} x - \pi & x > 1 \end{cases}$$



$$\text{in } (0, 1), f(x) + g(x) + h(x) = \frac{\pi}{2}$$

$$2 \tan^{-1} x + 2 \tan^{-1} x + 2 \tan^{-1} x = \frac{\pi}{2} \Rightarrow \tan^{-1} x = \frac{\pi}{12} \Rightarrow x = \tan\left(\frac{\pi}{12}\right) = 2 - \sqrt{3} \text{ Ans.}$$

$$\text{in } (-1, 0), 2 \tan^{-1} x - 2 \tan^{-1} x + 2 \tan^{-1} x = \frac{\pi}{2} \Rightarrow \tan^{-1} x = \frac{\pi}{4} \Rightarrow x = \tan\left(\frac{\pi}{4}\right) = 1$$

25.  $f(2) + g(2) + h(2)$  has the value equal to :

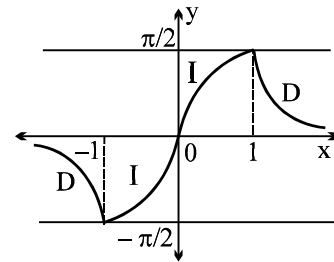
$f(2) + g(2) + h(2)$  का मान होगा :

(A)  $\cot^{-1}\left(\frac{-4}{3}\right)$  (B)  $\cot^{-1}\left(\frac{-3}{4}\right)$  (C)  $\pi - \tan^{-1}\left(\frac{2}{3}\right)$  (D)  $\cos^{-1}\left(\frac{-3}{4}\right)$

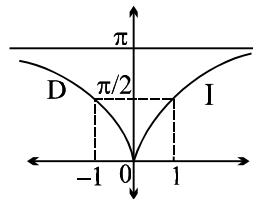
**Ans.** B

**Sol.** starting with  $x = \tan \theta$

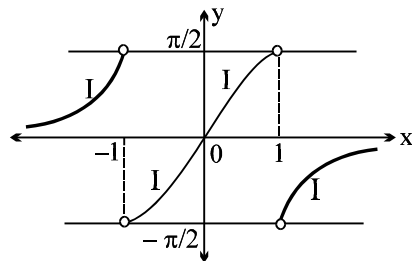
$$f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \begin{cases} 2 \tan^{-1} x & -1 \leq x \leq 1 \\ \pi - 2 \tan^{-1} x & \text{if } x \geq 1 \\ -\pi - 2 \tan^{-1} x & x \leq -1 \end{cases}$$



$$g(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \begin{cases} 2 \tan^{-1} x & x \geq 0 \\ -2 \tan^{-1} x & x < 0 \end{cases}$$



$$h(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right) = \begin{cases} \pi + 2 \tan^{-1} x & x < -1 \\ 2 \tan^{-1} x & -1 < x < 1 \\ 2 \tan^{-1} x - \pi & x > 1 \end{cases}$$



Obvious from the graphs.  $f(x) \rightarrow 2$ ;  $g(x) \rightarrow 1$ ;  $h(x) \rightarrow 0 \Rightarrow 3$  critical points. Ans.

**Paragraph for Question Nos. 26 to 28**

Given that  $\tan^{-1}\left(\frac{2x}{1-x^2}\right) = \begin{cases} 2 \tan^{-1} x & , \quad |x| < 1 \\ -\pi + 2 \tan^{-1} x & , \quad x > 1 \\ \pi + 2 \tan^{-1} x & , \quad x < -1 \end{cases}$

$$\sin^{-1} \frac{2x}{1+x^2} = \begin{cases} 2 \tan^{-1} x & \text{if } |x| \leq 1 \\ \pi - 2 \tan^{-1} x & \text{if } x > 1 \\ -(\pi + 2 \tan^{-1} x) & \text{if } x < -1 \end{cases}$$

and  $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$  for  $-1 \leq x \leq 1$

दिया गया है कि  $\tan^{-1}\left(\frac{2x}{1-x^2}\right) = \begin{cases} 2 \tan^{-1} x & , \quad |x| < 1 \\ -\pi + 2 \tan^{-1} x & , \quad x > 1 \\ \pi + 2 \tan^{-1} x & , \quad x < -1 \end{cases}$

$$\sin^{-1} \frac{2x}{1+x^2} = \begin{cases} 2 \tan^{-1} x & \text{यदि } |x| \leq 1 \\ \pi - 2 \tan^{-1} x & \text{यदि } x > 1 \\ -(\pi + 2 \tan^{-1} x) & \text{यदि } x < -1 \end{cases}$$

तथा  $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$ , जहाँ  $-1 \leq x \leq 1$

26.  $\sin^{-1}\left(\frac{4x}{x^2+4}\right) + 2 \tan^{-1}\left(-\frac{x}{2}\right)$  is independent from x then :

$\sin^{-1}\left(\frac{4x}{x^2+4}\right) + 2 \tan^{-1}\left(-\frac{x}{2}\right)$ , x से स्वतंत्र है, तो :

(A)  $x \in [-3, 4]$       (B)  $x \in [-2, 2]$       (C)  $x \in [-1, 1]$       (D)  $x \in [1, \infty]$

**Ans.** B

**Sol.**  $\sin^{-1}\left(\frac{4x}{x^2+4}\right) + 2 \tan^{-1}\left(-\frac{x}{2}\right)$

$$f(x) = \sin^{-1}\left(\frac{2\left(\frac{x}{2}\right)}{1+\left(\frac{x}{2}\right)^2}\right) - 2 \tan^{-1}\left(\frac{x}{2}\right)$$

$$\frac{x}{2} = \tan \theta$$

$$= \sin^{-1}(\sin(2\theta)) - 2 \tan^{-1}(\tan \theta)$$

$$\left| \frac{x}{2} \right| \leq 1$$

$$-1 \leq \frac{x}{2} \leq 1$$

$$-2 \leq x \leq 2$$

27. If  $\cos^{-1} \frac{6x}{1+9x^2} = -\frac{\pi}{2} + 2 \tan^{-1} 3x$ , then  $x \in$

- (A)  $\left(\frac{1}{3}, \infty\right)$  (B)  $(-1, \infty)$  (C)  $(-\infty, -1)$  (D) None of these

यदि  $\cos^{-1} \frac{6x}{1+9x^2} = -\frac{\pi}{2} + 2 \tan^{-1} 3x$ , तो  $x \in$

- (A)  $\left(\frac{1}{3}, \infty\right)$  (B)  $(-1, \infty)$  (C)  $(-\infty, -1)$  (D) इनमें से कोई नहीं

**Ans.** A

**Sol.**  $\cos^{-1} \frac{6x}{1+9x^2} = -\frac{\pi}{2} + 2 \tan^{-1} 3x$

$$\cos^{-1} \left( \frac{3x+3x}{1+(3x)(3x)} \right) = -\frac{\pi}{2} + 2 \tan^{-1}(3x)$$

$$\Rightarrow 3x = \tan \theta \quad |3x| \leq 1 \Rightarrow |x| \leq \frac{1}{3}$$

$$\cos^{-1} \left( \frac{2 \tan \theta}{1 + \tan^2 \theta} \right)$$

$$\Rightarrow \cos^{-1}(\sin 2\theta) = \frac{\pi}{2} - \sin^{-1}(\sin 2\theta)$$

$$\Rightarrow \frac{\pi}{2} - 2\theta$$

$$\frac{\pi}{2} - 2 \tan^{-1} 3x + \frac{\pi}{2} = 2 \tan^{-1}(3x)$$

$$4 \tan^{-1}(3x) = \pi$$

$$\tan^{-1}(3x) = \frac{\pi}{4}$$

$$3x = 1$$

$$x = \frac{1}{3}$$

$$\text{where } 3x > 1 \Rightarrow x > \frac{1}{3} \Rightarrow x \in \left(\frac{1}{3}, \infty\right) \quad \dots\dots\dots(2)$$

$$\pi = \pi$$

$$\text{where } 3x < 1 \Rightarrow x < \frac{1}{3} \Rightarrow x \in \left(-\infty, \frac{1}{3}\right) \quad \dots\dots\dots(3)$$

$$(\pi + 2 \tan^{-1} x) + 2 \tan^{-1} x \neq \pi$$

28. If  $(x-1)(x^2+1) > 0$ , then  $\sin\left(\frac{1}{2} \tan^{-1} \frac{2x}{1-x^2} - \tan^{-1} x\right) =$

- (A) 1                      (B)  $\frac{1}{\sqrt{2}}$                       (C) -1                      (D) None of these

यदि  $(x-1)(x^2+1) > 0$ , तो  $\sin\left(\frac{1}{2} \tan^{-1} \frac{2x}{1-x^2} - \tan^{-1} x\right) =$

- (A) 1                      (B)  $\frac{1}{\sqrt{2}}$                       (C) -1                      (D) इनमें से कोई नहीं

**Ans.** C

**Sol.**  $(x-1)(x^2+1) > 0 \Rightarrow x > 1$

$$\sin\left(\frac{1}{2} \tan^{-1} \frac{2x}{1-x^2} - \tan^{-1} x\right)$$

$$\tan^{-1} x = \theta$$

$$x = \tan \theta$$

$$\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

$$\sin^{-1}\left(\frac{2x}{1-x^2}\right) = \tan^{-1}(\tan 2\theta) \Rightarrow (2\theta - \pi); 2\theta \in \left(\frac{\pi}{2}, \pi\right)$$

$$\tan^{-1}(\tan \theta); \theta - 0; \theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

$$\sin^{-1}\left[\frac{2\theta - \pi}{2} - \theta\right]$$

$$\sin^{-1}\left[\theta - \frac{\pi}{2} - \theta\right]$$

$$\Rightarrow -1$$

### ONE OR MORE THAN ONE CORRECT TYPE

29. In the following functions defined from  $[-1, 1]$  to  $[-1, 1]$  the functions which are not bijective are :

अन्तराल  $[-1, 1]$  से  $[-1, 1]$  में परिभाषित कौनसे फलन एकैकी आच्छादक नहीं होंगे :

- (A)  $\sin(\sin^{-1}x)$       (B)  $\frac{2}{\pi}\sin^{-1}(\sin x)$       (C)  $(\operatorname{sgn} x) \ln e^x$       (D)  $x^3 \operatorname{sgn} x$

**Ans.** BCD

**Sol.**  $f: [-1, 1] \rightarrow [-1, 1]$

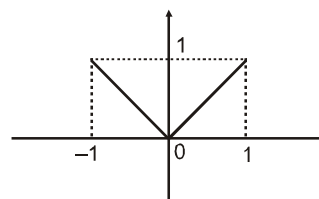
(A)  $f(x) = \sin(\sin^{-1}x) = x, \quad x \in [-1, 1]$       Bijjective function एकैकी आच्छादक फलन  $y \in [-1, 1]$

(B)  $f(x) = \frac{2}{\pi}\sin^{-1}(\sin x) = \frac{2x}{\pi}, \quad x \in [-1, 1]$

Not bijective एकैकी आच्छादक फलन नहीं है।  $y \in \left[\frac{-2}{\pi}, \frac{2}{\pi}\right]$

(C)  $f(x) = (\operatorname{sgn} x)(\ln e^x)$

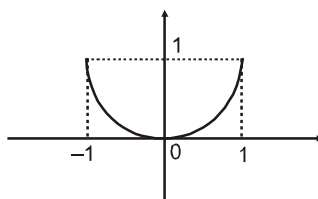
$$= \begin{cases} x & , \quad x \in (0, 1] \\ 0 & , \quad x = 0 \\ -x & , \quad x \in [-1, 0) \end{cases}$$



Not bijective एकैकी आच्छादक फलन नहीं है।

(D)  $f(x) = x^3 \operatorname{sgn} x$

$$= \begin{cases} x^3 & , \quad x > 0 \\ 0 & , \quad x = 0 \\ -x^3 & , \quad x < 0 \end{cases}$$



Not bijective एकैकी आच्छादक फलन नहीं है।

30. If  $\alpha, \beta, \gamma$  are three angles given by  $\alpha = 2 \tan^{-1}(\sqrt{2} - 1)$ ,  $\beta = 3 \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$  and  $\gamma = \cos^{-1} \frac{1}{3}$ , then:

यदि  $\alpha, \beta, \gamma$  तीन कोण निम्न प्रकार दिए गए  $\alpha = 2 \tan^{-1}(\sqrt{2} - 1)$ ,  $\beta = 3 \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$  तथा  $\gamma = \cos^{-1} \frac{1}{3}$ ,

तब :

- (A)  $\alpha > \beta$  (B)  $\beta > \gamma$  (C)  $\alpha < \gamma$  (D)  $\beta < \gamma$

**Ans.** BC

**Sol.**  $\alpha = 2 \tan^{-1}(\sqrt{2} - 1) = 2 \times 15 = 30 = \frac{\pi}{6} = \frac{2\pi}{12}$

$$\beta = 3 \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$$

$$= 3 \times \frac{\pi}{4} - \frac{\pi}{6} = \frac{7\pi}{12}$$

$$\gamma = \cos^{-1}\left(\frac{1}{3}\right)$$

$$\frac{\pi}{3} < \gamma < \frac{\pi}{2}$$

$$\frac{4\pi}{12} < \gamma < \frac{6\pi}{12}$$

31. Which of the following are identical functions? (where  $[x]$  denotes greatest integer less than or equal  $x$ ,  $\{x\}$  denotes fractional part of  $x$  and  $\text{sgn } x$  denotes signum function of  $x$  respectively.)

निम्न में से कौन से फलन समान हैं? (जहाँ  $[x]$ , महत्तम पूर्णांक को निरूपित करता है जो  $x$  अथवा  $x$  से छोटा है एवं  $\{x\}$ ,  $x$  के भिन्नात्मक भाग को तथा  $\text{sgn } x$  सिग्नम फलन को दर्शाता है)

(A)  $f(x) = \text{sgn}(|x| + 1)$  (B)  $g(x) = \sin^2(\ln x) + \cos^2(\ln x)$

(C)  $h(x) = \frac{2}{\pi}(\sin^{-1}\{x\} + \cos^{-1}\{x\})$  (D)  $k(x) = \sec^2[x] - \tan^2\{x\}$

**Ans.** ACD

**Sol.** (A)  $\text{sgn}(|x| + 1) = 1 \forall x \in \mathbb{R}$ .

(B)  $\sin^2(\ln x) + \cos^2(\ln x) = 1 \forall x \in \mathbb{R}^+$

$$(C) \frac{2}{\pi} (\sin^{-1}\{x\} + \cos^{-1}\{x\}) = 1 \quad \forall x \in \mathbb{R}.$$

$$(D) \sec^2 [\{x\}] - \tan^2 \{\{x\}\} = \sec^2 0^\circ - \tan^2 0^\circ = 1 \quad \forall x \in \mathbb{R}.$$

32. The value of  $a$  for which  $ax + \sec^{-1} \sqrt{2x^2 - x^4} + \operatorname{cosec}^{-1} \sqrt{2x^2 - x^4} = 0$  :

$a$  का मान ज्ञात कीजिये जिसके लिये  $ax + \sec^{-1} \sqrt{2x^2 - x^4} + \operatorname{cosec}^{-1} \sqrt{2x^2 - x^4} = 0$  :

$$(A) \frac{\pi}{2} \quad (B) -\frac{\pi}{2} \quad (C) \frac{2}{\pi} \quad (D) -\frac{2}{\pi}$$

**Ans.** AB

**Sol.** for existence of  $\sec^{-1}(\sqrt{2x^2 - x^4})$  or  $\operatorname{cosec}^{-1}(\sqrt{2x^2 - x^4})$

$$2x^2 - x^4 \geq 1$$

$$\text{i.e. } (x^2 - 1)^2 \leq 0 \quad \text{hence } x^2 = 1 \Rightarrow x = \pm 1$$

$$\text{if } x = 1, a + \frac{\pi}{2} = 0;$$

$$\text{if } x = -1, -a + \frac{\pi}{2} = 0 \Rightarrow a = \pm \frac{\pi}{2} \text{ Ans.}$$

33. Let function  $f(x)$  be defined as  $f(x) = |\sin^{-1}x| + \cos^{-1}\left(\frac{1}{x}\right)$ . Then which of the following is/are TRUE?

(A)  $f(x)$  is injective in its domain

(B)  $f(x)$  is many-one in its domain

(C) Range of  $f$  is a singleton set

(D)  $\operatorname{sgn}(f(x)) = 1$  where  $\operatorname{sgn} x$  denotes signum function of  $x$

माना फलन  $f(x)$  परिभाषित है  $f(x) = |\sin^{-1}x| + \cos^{-1}\left(\frac{1}{x}\right)$  तो निम्न में से कौनसा/कौनसे कथन सत्य है?

(A)  $f(x)$  स्वयं की प्रान्त में एकैकी है

(B)  $f(x)$  स्वयं की प्रान्त में बहुएकैकी है

(C)  $f(x)$  का परिसर एकल समुच्चय है

(D)  $\operatorname{sgn}(f(x)) = 1$  जहाँ  $\operatorname{sgn} x$ ,  $x$  का सिग्नम फलन है

**Ans.** AD

$$\text{Sol. } f(x) = |\sin^{-1}x| + \cos^{-1}\left(\frac{1}{x}\right)$$

Domain of  $f(x)$  is  $\{-1, 1\}$

$$f(1) = \frac{\pi}{2}, f(-1) = \frac{3\pi}{2}$$

So function  $f(x)$  is injective

$$\operatorname{sgn}(f(x)) = 1 \quad (f(x) > 0)$$

$$\text{Range of } f(x) = \left\{ \frac{\pi}{2}, \frac{3\pi}{2} \right\}$$

34. Which of the following statement(s) is/are TRUE?

- (A) Domain of  $y = \cos^{-1}(e^x)$  is same as range of  $y = -\sqrt{-x}$   
 (B) Number of elements common in the range of function  $y = \tan^{-1}(\operatorname{sgn} x)$  and  $y = \cot^{-1}(\operatorname{sgn} x)$  is only 1 (where  $\operatorname{sgn} x$  denotes signum function of  $x$ )  
 (C) The function  $y = \operatorname{sgn}(\cot^{-1} x)$  and  $y = 1$  are identical functions  
 (D) Number of integers in the solution set of  $1 < \log_2(\tan^{-1} x) < 2$  is 4

निम्न में से कौन कौन से कथन सत्य हैं?

- (A)  $y = \cos^{-1}(e^x)$  का प्रान्त तथा  $y = -\sqrt{-x}$  का परिसर समान है  
 (B)  $y = \tan^{-1}(\operatorname{sgn} x)$  और  $y = \cot^{-1}(\operatorname{sgn} x)$  के परिसर में उभयनिष्ठ अवयवों की संख्या 1 है  
 (C)  $y = \operatorname{sgn}(\cot^{-1} x)$  और  $y = 1$  समान फलन है  
 (D)  $1 < \log_2(\tan^{-1} x) < 2$  के हल समुच्चय में पूर्णाकों की संख्या 4 है

**Ans.** ABC

**Sol.** (A) For domain of  $y = \cos^{-1}(e^x)$ ,  $0 < e^x \leq 1 \Rightarrow x \leq 0$  or  $x \in (-\infty, 0]$

$$\text{For range of } y = -\sqrt{-x}, \quad -y = \sqrt{-x}$$

$$\sqrt{-x} \geq 0 \Rightarrow -y \geq 0 \Rightarrow y \in (-\infty, 0]$$

$\therefore$  Domain of  $y = \cos^{-1}(e^x)$  is identical with range of  $y = -\sqrt{-x}$ .

$$(B) \quad y = \tan^{-1}(\operatorname{sgn} x) = \begin{cases} \tan^{-1}(-1), & \forall x < 0 \\ \tan^{-1} 0, & \text{for } x = 0 \\ \tan^{-1}(1), & \forall x > 0 \end{cases} \Rightarrow \text{Range of } y = \tan^{-1}(\operatorname{sgn} x) \text{ is } \left\{ -\frac{\pi}{4}, 0, \frac{\pi}{4} \right\}$$

$$\text{Also, } y = \cot^{-1}(\operatorname{sgn} x) = \frac{\pi}{2} - \tan^{-1}(\operatorname{sgn} x) \Rightarrow \text{Range of } y = \cot^{-1}(\operatorname{sgn} x) \text{ is } \left\{ \frac{3\pi}{4}, \frac{\pi}{2}, \frac{\pi}{4} \right\}$$

$\therefore$  Number of elements common in the range of  $y = \tan^{-1}(\operatorname{sgn} x)$  and  $y = \cot^{-1}(\operatorname{sgn} x)$  is only one i.e.  $\frac{\pi}{4}$ .

$$(C) \quad y = \operatorname{sgn}(\cot^{-1} x) = 1 \quad \forall x \in \mathbb{R}$$

$\therefore y = \operatorname{sgn}(\cot^{-1} x)$  and  $y = 1$  are identical functions.

$$(D) \quad 1 < \log_2(\tan^{-1} x) < 2 \Rightarrow 2^1 < \tan^{-1} x < 2^2 \Rightarrow \text{Not possible.} \quad \left( \text{As } \frac{-\pi}{2} < \tan^{-1} x < \frac{\pi}{2} < 2 \right)$$



35.  $2\cot^{-1} 7 + \cos^{-1} \frac{3}{5}$  is equal to :

$2\cot^{-1} 7 + \cos^{-1} \frac{3}{5}$  का मान होगा :

- (A)  $\cot^{-1} \frac{44}{117}$       (B)  $\operatorname{cosec}^{-1} \frac{125}{117}$       (C)  $\tan^{-1} \frac{4}{117}$       (D)  $\cos^{-1} \frac{44}{125}$

**Ans.** ABD

**Sol.**  $2\cot^{-1} 7 + \cos^{-1} \frac{3}{5}$

$$2 \tan^{-1} \left( \frac{1}{7} \right) + \tan^{-1} \left( \frac{4}{3} \right)$$

$$\Rightarrow \tan^{-1} \left( \frac{\frac{2}{7}}{1 - \frac{1}{49}} \right) + \tan^{-1} \left( \frac{4}{3} \right)$$

$$\Rightarrow \tan^{-1} \left( \frac{2 \times 7}{48} \right) + \tan^{-1} \left( \frac{4}{3} \right)$$

$$\Rightarrow \tan^{-1} \left( \frac{\frac{7}{24} + \frac{4}{3}}{1 - \frac{7}{24} \times \frac{4}{3}} \right)$$

$$\Rightarrow \tan^{-1} \left( \frac{\frac{7+32}{24}}{\frac{18-7}{18}} \right)$$

$$\Rightarrow \tan^{-1} \left( \frac{39}{4 \times 11} \times 3 \right)$$

$$\tan^{-1} \left( \frac{117}{44} \right)$$

36. If  $\tan^2(\tan^{-1}x) = \operatorname{cosec}^{-1}(\operatorname{cosec}7) + \sec(\sec^{-1}8) + \cot^{-1}(\cot(-6))$ , then  $\tan^{-1}(\tan(18 + x^2))$  :

- (A) is always positive      (B) is always negative      (C) is irrational      (D) is rational

यदि  $\tan^2(\tan^{-1}x) = \operatorname{cosec}^{-1}(\operatorname{cosec}7) + \sec(\sec^{-1}8) + \cot^{-1}(\cot(-6))$ , तो  $\tan^{-1}(\tan(18 + x^2))$  :

(A) सदैव धनात्मक होगा (B) सदैव ऋणात्मक होगा (C) अपरिमेय होगा (D) परिमेय होगा

**Ans.** BC

**Sol.**  $\tan^2(\tan^{-1}x) = \operatorname{cosec}^{-1}(\operatorname{cosec}7) + \sec(\sec^{-1}8) + \cot^{-1}(\cot(-6))$

$$x^2 = 7 - 2\pi + 8 - 2\pi - 6 - (-6)\pi$$

$$x^2 = 9 + 2\pi$$

$$\tan^{-1} \tan(18 + x^2) = \tan^{-1}(\tan(18 + 9 + 2\pi))$$

$$\Rightarrow \tan^{-1}(\tan 27)$$

$$\Rightarrow 27 - 9\pi$$

37. Let  $\tan^{-1}\left(\tan \frac{5\pi}{4}\right) = \alpha$ ,  $\tan^{-1}\left(-\tan \frac{2\pi}{3}\right) = \beta$ . Then :

माना  $\tan^{-1}\left(\tan \frac{5\pi}{4}\right) = \alpha$ ,  $\tan^{-1}\left(-\tan \frac{2\pi}{3}\right) = \beta$  है, तो :

(A)  $\alpha > \beta$  (B)  $4\alpha - 3\beta = 0$  (C)  $\alpha + \beta = \frac{7\pi}{12}$  (D)  $\alpha + \beta = \pi$

**Ans.** BC

**Sol.**  $\tan^{-1} \tan\left(\frac{5\pi}{4}\right) = \alpha$

$$\alpha = \tan^{-1} \tan\left(\frac{\pi}{4}\right)$$

$$\alpha = \frac{\pi}{4}$$

$$\beta = \tan^{-1}\left(-\tan \frac{2\pi}{3}\right)$$

$$\beta = -\tan^{-1}\left(\tan\left(\pi - \frac{\pi}{3}\right)\right)$$

$$\beta = +\tan^{-1}\left(\tan \frac{\pi}{3}\right)$$

$$\beta = +\frac{\pi}{3}$$

38.  $2 \tan(\tan^{-1}(x) + \tan^{-1}(x^3))$  where  $x \in \mathbb{R} - \{-1, 1\}$  is equal to :

$2 \tan(\tan^{-1}(x) + \tan^{-1}(x^3))$  जहाँ  $x \in \mathbb{R} - \{-1, 1\}$  बराबर होगा :

(A)  $\frac{2x}{1-x^2}$  (B)  $\tan(2 \tan^{-1}x)$

(C)  $\tan(\cot^{-1}(-x) - \cot^{-1}(x))$  (D)  $\tan(2 \cot^{-1}x)$

**Ans.** ABC

**Sol.**  $2 \tan(\tan^{-1}(x) + \tan^{-1}(x^3))$

$$\Rightarrow 2 \tan\left(\tan^{-1}\left(\frac{x+x^3}{1-x^4}\right)\right)$$

$$= 2 \tan\left[\tan^{-1}\left(\frac{x(x^2+1)}{(1-x^2)(1+x^2)}\right)\right]$$

$$= 2 \tan\left[\tan^{-1}\left(\frac{x}{1-x^2}\right)\right]$$

$$= \frac{2x}{1-x^2}$$

(D)  $\tan(2 \cot^{-1}x)$   $\cot^{-1} x = \theta; \theta \in (0, \pi)$

$$\cot \theta = x$$

$$\tan \theta = \frac{1}{x}$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{\frac{2}{x}}{1 - \frac{1}{x^2}} = \frac{2x}{x^2 - 1}$$

(C)  $\tan(\cot^{-1}(-x) - \cot^{-1} x)$

$$\Rightarrow \tan(\pi - 2 \cot^{-1} x)$$

$$= -\tan(2 \cot^{-1} x)$$

$$\Rightarrow \cot^{-1} x = \theta$$

$$= \tan 2\theta$$

$$= \frac{-2 \tan \theta}{1 - \tan^2 \theta} = \frac{-\frac{2}{x}}{1 - \frac{1}{x^2}} = \frac{2x}{1 - x^2}$$

(B)  $\tan(2 \tan^{-1} x)$

$$\Rightarrow \tan^{-1} x = \theta$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{2x}{1 - x^2}$$

39. For the equation  $2x = \tan(2 \tan^{-1} a) + 2 \tan(\tan^{-1} a + \tan^{-1} a^3)$ , which of the following is invalid?

समीकरण  $2x = \tan(2 \tan^{-1} a) + 2 \tan(\tan^{-1} a + \tan^{-1} a^3)$  के लिए निम्न में से कौन सा असत्य है?

(A)  $a^2 x + 2a = x$  (B)  $a^2 + 2ax + 1 = 0$  (C)  $a \neq 0$  (D)  $a \neq -1, 1$

**Ans.** BC

**Sol.**  $2x = \tan(2 \tan^{-1} a) + 2 \tan(\tan^{-1} a + \tan^{-1} a^3)$

$$2x = \tan \left( \tan^{-1} \left( \frac{2a}{1 - a^2} \right) \right) + 2 \tan \left( \tan^{-1} \left( \frac{a + a^3}{1 - a^4} \right) \right)$$

$$= \frac{2a}{1 - a^2} + \frac{a}{1 - a^2}$$

$$x = \frac{2a}{1 - a^2}$$

$$x - xa^2 = 2a$$

$$xa^2 + 2a = x$$

$$a \neq \pm 1$$

40. If integral pair  $(x, y)$  satisfy the equation  $\arctan \frac{1}{x} + \arctan \frac{1}{y} = \arctan \frac{1}{10}$ , then sum of  $x + y$  can be :

यदि पूर्णांक युग्म  $(x, y)$  समीकरण  $\arctan \frac{1}{x} + \arctan \frac{1}{y} = \arctan \frac{1}{10}$  का संतुष्ट करते हैं तो  $x + y$  का मान हो सकता/सकते

है:

(A) 122 (B) 100 (C) 92 (D) -92

**Ans.** A

**Sol.**  $\tan^{-1}\left(\frac{1}{x}\right) + \tan^{-1}\left(\frac{1}{y}\right) = \tan^{-1}\left(\frac{1}{10}\right)$

$$= \tan^{-1}\left(\frac{\frac{1}{x} + \frac{1}{y}}{1 - \frac{1}{xy}}\right)$$

$$\tan^{-1}\left(\frac{x+y}{2y-1}\right) = \tan^{-1}\left(\frac{1}{10}\right)$$

$$\frac{x+y}{xy-1} = \frac{1}{10}$$

$$10x + 10y = xy - 1$$

$$10x - xy + 10y + 1 = 0$$

$$10x + 10y - xy + 1 = 0$$

$$x(10 + y) + 10y - y = -y$$

$$x(10 - y) + 10y - 10y - 100 = -101$$

$$x(10 - y) - 10y - 10y - 100 = -101$$

$$(x - 10) - 10(10 - y) = 101$$

$$(x - 10)(y - 10) = 101$$

$$x - 10 = 101 \quad y - 10 = 1$$

$$x - 10 = 1$$

$$y - 10 = 101$$

$$x = 111 \quad y = 11$$

$$x = 11$$

$$y = 111$$

$$x - 10 = -1, \quad y - 10 = -101$$

$$x - 10 = -101$$

$$y - 10 = -1$$

$$x = 9 \quad y = -91$$

$$x = -91$$

$$y = 9$$

### MATCH THE COLUMN

41. Match the following column :

#### Column-I

#### Column-II

(A) Period of  $\frac{1}{2} \left\{ \frac{|\sin x|}{\cos x} + \frac{|\cos x|}{\sin x} \right\}$

(P) 2

(B) Range of  $\sin^{-1} \sqrt{\log_{[x]} \frac{|x|}{x}}$

(Q)  $2\pi$

(C) Total number of solution  $x^2 - 4 - [x] = 0$

(R) 1

(D) Period of  $f(x) = \sin^2 x + \cos x$

(S)  $\{0\}$

[where  $[ ]$  denotes G.I.F]

निम्न कॉलम को सुमेलित कीजिए :

**Column-I**

**Column-II**

(A)  $\frac{1}{2} \left\{ \frac{|\sin x|}{\cos x} + \frac{|\cos x|}{\sin x} \right\}$  का आवर्तकाल होगा

(P) 2

(B)  $\sin^{-1} \sqrt{\log_{[x]} \frac{|x|}{x}}$  का परिसर होगा

(Q)  $2\pi$

(C)  $x^2 - 4 - [x] = 0$  के कुल हल होंगे

(R) 1

(D)  $f(x) = \sin^2 x + \cos x$  का आवर्तकाल होगा

(S)  $\{0\}$

[जहाँ  $[ ]$  महत्तम पूर्णांक फलन है]

**Ans.** (A) Q (B) S (C) P (D) Q

**Sol.** (A)  $\frac{1}{2} \left\{ \frac{|\sin x|}{\cos x} + \frac{|\cos x|}{\sin x} \right\}$

$$P = 2\pi \quad P = \pi$$

$$\text{LCM } \{2\pi, \pi\} = 2\pi$$

(B)  $\sin^{-1} \sqrt{\log_{[x]} \frac{|x|}{x}}$

$$\frac{|x|}{x} > 0 \quad [x] > 0, x > 0$$

$$x > 1, x > 2$$

$$\sin^{-1} \sqrt{\log_{[x]}(1)}$$

$$\sin^{-1} \sqrt{0} = 0$$

$$(C) x^2 - 4 - [x] = 0$$

$$x^2 = 4 - (x - \{x\}) = 6$$

$$x^2 - x - 4 + \{x\} = 0$$

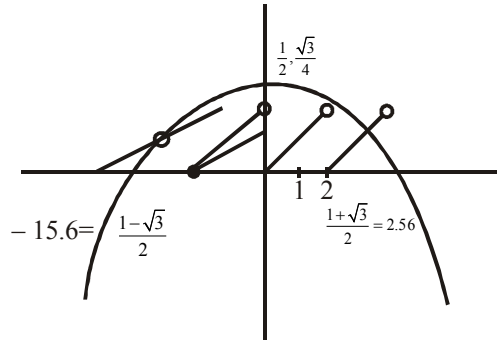
$$\{x\} = -x^2 + x + 4$$

$$[0, 1)$$

$$-x^2 + x + 4 = 0$$

$$x = \frac{-1 \pm \sqrt{1-16}}{-2}$$

$$x = \frac{1 \pm \sqrt{3}}{2}$$



$$(D) f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$$

$$\text{LCM}(\pi, 2\pi) = 2\pi$$

42. Match the following column :

#### Column-I

$$(A) f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$$

$$(B) g(x) = \sin^{-1}|x| + 2 \tan^{-1}|x|$$

$$(C) h(x) = 2\sin^{-1}\left(\frac{2x}{1+x^2}\right), x \in [0, 1]$$

$$(D) k(x) = \cot(\cot^{-1}x)$$

निम्न कॉलम को सुमेलित कीजिए :

#### Column-I

$$(A) f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$$

$$(B) g(x) = \sin^{-1}|x| + 2 \tan^{-1}|x|$$

$$(C) h(x) = 2\sin^{-1}\left(\frac{2x}{1+x^2}\right), x \in [0, 1]$$

$$(D) k(x) = \cot(\cot^{-1}x)$$

Ans. (A) PQR (B) PQ (C) PQ (D) Q

Sol. (A)  $f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$

$$2\pi \quad 2\pi$$

$$\text{LCM}(\pi, 2\pi) = 2\pi$$

$$f'(x) = \frac{\cos x}{\sqrt{1-\sin^2 x}} + \frac{-1-\sin x}{\sqrt{1-\cos^2 x}}$$

$$= \frac{\cos x}{\sqrt{1-\sin^2 x}} + \frac{\sin x}{\sqrt{1-\cos^2 x}}$$

$$f'(x) > 0 \text{ in } x \in (0, 1)$$

#### Column-II

(P) range is  $[0, \pi]$

(Q) is increasing  $\forall x \in (0, 1)$

(R) period is  $2\pi$

(S) is decreasing  $\forall x \in (0, 1)$

#### Column-II

(P) परिसर  $[0, \pi]$  है

(Q)  $\forall x \in (0, 1)$  के लिए वर्धमान

(R) आवृत्त काल  $2\pi$  है

(S)  $\forall x \in (0, 1)$  के लिए ह्रसमान है

$$f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\cos x)$$

↓

↓

$$x \in \left[ \frac{\pi}{2}, \frac{\pi}{2} \right] \quad x \in [0, \pi]$$

$$x \in \left[ 0, \frac{\pi}{2} \right] \Rightarrow \text{Range} \Rightarrow \left[ 0, \frac{\pi}{2} \right] + \left[ 0, \frac{\pi}{2} \right] = [0, \pi]$$

$$(B) g(x) = \sin^{-1}|x| + 2 \tan^{-1}|x|$$

$$|x| \leq 1$$

$$\text{Range} \Rightarrow \left[ 0, \frac{\pi}{2} \right] + 2 \left[ 0, \frac{\pi}{4} \right]$$

$$\Rightarrow [0, \pi]$$

$$f(x) = 3x \Rightarrow \text{increasing}$$

$$(C) h(x) = 2 \sin^{-1} \left( \frac{2x}{1+x^2} \right), \quad x \in [0, 1]$$

$$x = \tan \theta \quad \theta \in \left[ 0, \frac{\pi}{4} \right]$$

$$= 2 \sin^{-1}(\sin 2\theta) \quad 2\theta \in \left[ 0, \frac{\pi}{2} \right]$$

$$\Rightarrow 2(2\theta)$$

$$= 4\theta$$

$$h(x) = \tan^{-1}(x) \text{ is increasing in } x \in (0, 1)$$

$$x \in [0, 1]$$

$$\text{Range} \Rightarrow [0, \pi]$$

$$(D) k(x) = \cot(\cot^{-1}x) \text{ increasing in } (0, 1)$$

$$\text{Range} = (0, \pi)$$

43. If  $S_n = \sum_{r=1}^n r!$  then for  $n > 6$   $\left( \text{given } \sum_{r=1}^6 r! = 873 \right)$  : (where  $[ ]$  denotes greatest integer function)



यदि  $S_n = \sum_{r=1}^n r!$  सभी  $n > 6$  के लिये  $\left( \sum_{r=1}^6 r! = 873 \right)$  (जहाँ  $[ ]$  महत्तम पूर्णांक फलन को दर्शाता है) सही स्तम्भों का मिलान कीजिए।

**Column-I**

(A)  $\sin^{-1} \left( \sin \left( S_n - 7 \left[ \frac{S_n}{7} \right] \right) \right)$

(B)  $\cos^{-1} \left( \cos \left( S_n - 7 \left[ \frac{S_n}{7} \right] \right) \right)$

(C)  $\tan^{-1} \left( \tan \left( S_n - 7 \left[ \frac{S_n}{7} \right] \right) \right)$

(D)  $\cot^{-1} \left( \cot \left( S_n - 7 \left[ \frac{S_n}{7} \right] \right) \right)$

**Column-II**

(P)  $5 - 2\pi$

(Q)  $2\pi - 5$

(R)  $6 - 2\pi$

(S)  $5 - \pi$

(T)  $\pi - 4$

**Ans.** (A) P (B) Q (C) P (D) S

**Sol.** (A)  $S_n = \sum_{r=1}^n r!$

$$\sin^{-1} \left( \sin \left( S_n - 7 \left[ \frac{S_n}{7} \right] \right) \right)$$

$$S_n = 1! + 2! + 3! + \dots + 6! + 7! + \dots + n!$$

$$S_n = 873 + 7! + 8! + \dots + n!$$

$$S_n = 873 + 7k$$

$$\left[ \frac{S_n}{7} \right] = \left[ \frac{873}{7} + \frac{7k}{7} \right]$$

$$\Rightarrow \left[ \frac{873}{7} \right] + k$$

$$\Rightarrow 124 + k$$

$$\Rightarrow 7 \left[ \frac{S_n}{7} \right] = 7(124 + k)$$

$$= 868 + 7k$$

$$\Rightarrow S_n - 7 \left[ \frac{S_n}{7} \right] = 873 + 7k - 868 - 7k$$

$$\Rightarrow 5$$

$$(A) \sin^{-1}(\sin 5) = 5 - \pi$$

$$(B) \cos^{-1}(\cos 5) = 2\pi - 5$$

$$(C) \tan^{-1}(\tan 5) = 5 - 2\pi$$

$$(D) \cot^{-1}(\cot 5) = 5 - \pi$$

44. Let  $(x, y)$  be such that  $\sin^{-1}(ax) + \cos^{-1}(y) + \cos^{-1}(bxy) = \frac{\pi}{2}$ .

Match the statements in Column I with statements in Column II and indicate your answer by darkening the appropriate bubbles in the  $4 \times 4$  matrix given in the ORS.

**Column-I**

(A) If  $a = 1$  and  $b = 0$ , then  $(x, y)$

(B) If  $a = 1$  and  $b = 1$ , then  $(x, y)$

(C) If  $a = 1$  and  $b = 2$ , then  $(x, y)$

(D) If  $a = 2$  and  $b = 2$ , then  $(x, y)$

**Column-II**

(P) lies on the circle  $x^2 + y^2 = 1$

(Q) lies on  $(x^2 - 1)(y^2 - 1) = 0$

(R) lies on  $y = x$

(S) lies on  $(4x^2 - 1)(y^2 - 1) = 0$

यदि  $(x, y)$  इस प्रकार है कि  $\sin^{-1}(ax) + \cos^{-1}(y) + \cos^{-1}(bxy) = \frac{\pi}{2}$  तो स्तम्भ-I का स्तम्भ-II से सही मिलान कीजिए व अपने उत्तरों को

**Column-I**

(A) यदि  $a = 1$  व  $b = 0$ , तो  $(x, y)$

(B) यदि  $a = 1$  व  $b = 1$ , तो  $(x, y)$

(C) यदि  $a = 1$  व  $b = 2$ , तो  $(x, y)$

(D) यदि  $a = 2$  व  $b = 2$ , तो  $(x, y)$

**Column-II**

(P) वृत्त  $x^2 + y^2 = 1$  पर स्थित है

(Q)  $(x^2 - 1)(y^2 - 1) = 0$  पर स्थित है

(R)  $y = x$  पर स्थित है

(S)  $(4x^2 - 1)(y^2 - 1) = 0$  पर स्थित है

**Ans.** (A) P (B) Q (C) P (D) S

**Sol.**  $\sin^{-1}(ax) + \cos^{-1}y + \cos^{-1}(bxy) = \frac{\pi}{2}$

$$(A) \quad a = 1, \quad b = 0 \Rightarrow \sin^{-1}(x) + \cos^{-1}(y) + \cos^{-1}(0) = \frac{\pi}{2} \Rightarrow \sin^{-1}x + \cos^{-1}y = 0$$

$$\Rightarrow \cos^{-1}y = -\sin^{-1}x \Rightarrow \cos^{-1}y = \cos^{-1}\sqrt{1-x^2} \Rightarrow x^2 + y^2 = 1$$

$$(B) \quad \sin^{-1}(x) + \cos^{-1}y + \cos^{-1}(xy) = \frac{\pi}{2} \Rightarrow \cos^{-1}(y) + \cos^{-1}(xy) = \cos^{-1}x.$$

$$\Rightarrow \cos^{-1}\left(xy^2 - \sqrt{(1-y^2)(1-x^2y^2)}\right) = \cos^{-1}x. \Rightarrow xy^2 - \sqrt{(1-y^2)(1-x^2y^2)} = x$$

$$\Rightarrow 1 - x^2 - y^2 + x^2y^2 = 0 \Rightarrow (1-x^2)(1-y^2) = 0$$

$$(C) \quad \sin^{-1}(x) + \cos^{-1}y + \cos^{-1}(2xy) = \frac{\pi}{2} \Rightarrow \cos^{-1}\left(2xy^2 - \sqrt{(1-y^2)(1-4x^2y^2)}\right) = \cos^{-1}x.$$

$$\Rightarrow 2xy^2 - \sqrt{(1-y^2)(1-4x^2y^2)} = x \Rightarrow 2xy^2 - x = \sqrt{(1-y^2)(1-4x^2y^2)}$$

$$\Rightarrow 4x^2y^4 + x^2 - 4x^2y^2 = 1 - y^2 - 4x^2y^2 + 4x^2y^4 \Rightarrow x^2 + y^2 = 1.$$

$$(D) \quad \sin^{-1}(2x) + \cos^{-1}y + \cos^{-1}(2xy) = \frac{\pi}{2} \Rightarrow \cos^{-1}\left(2y^2x - \sqrt{(1-y^2)(1-4x^2y^2)}\right) = \cos^{-1}(2x)$$

$$\Rightarrow 2y^2x - \sqrt{1-y^2-4x^2y^2+4x^2y^4} = 2x. \Rightarrow 1 - 4x^2 - y^2 + 4x^2y^2 = 0$$

$$\Rightarrow (1 - 4x^2)(1 - y^2) = 0.$$